Multicast Service Function Chain Orchestration in SDN/NFV-Enabled Networks: Embedding, Readjustment, and Expanding

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***Abstract*—Multicast is an effective transmission mode to support ever-growing multimedia applications. The introduction of software defined networking (SDN) and network function virtualization (NFV) makes the multicast service operation more flexible and efficient. Nevertheless, one main challenge of SDN/NFV-enabled multicast is optimally orchestrating the service function chain (SFC) to match service and network resources. Compared with unicast, multicast SFC orchestration (MSO) is more challenging due to the features of multicast service like multicast routing and user fluidity (i.e., frequent user arrival and departure). There are still some gaps in the joint optimization of MSO and multicast routing, and very little attention is paid to user fluidity. In this paper, we study the MSO in SDN/NFV-enabled networks encompassing multicast SFC embedding (MSE), multicast SFC readjustment (MSR), and multicast SFC expanding (MSEP), three types of MSO. For each kind of MSO, we simultaneously consider several key optimization factors when jointly optimizing MSO and multicast routing. Besides, aside from MSE, we investigate two new types of MSO: MSR and MSEP, for efficient orchestration under the fluidity of users. Specifically, we define and formulate MSE, MSR, and MSEP problems and develop three novel algorithms to respectively solve them. Simulation results demonstrate that our algorithms outperform benchmark algorithms and achieve near-optimal performance.**

***Index Terms*—Multicast, SDN/NFV-enabled network, service function chain (SFC), virtual network function (VNF).**

I. Introduction

RECENT years have witnessed a rapidly rising trend of multimedia applications. On the one hand, the reduction in video production and distribution costs has promoted the development of streaming media applications such as video streaming and live streaming. According to [1], the global video streaming market is expected to expand at a compound annual growth rate of 21.0% from 2022 to 2028. On the other hand, the ongoing coronavirus disease has forced people to use more online platforms for daily tasks, leading to an increasing demand for multimedia applications like video conferencing and online learning. Multicast is an effective one-to-many data transmission mode to support currently ever-growing multimedia applications. By replicating packets at necessary intermediate nodes and delivering copies to a group of different destinations, multicast can significantly reduce bandwidth consumption compared with unicast [2]. However, traditional network suffers from rigid architecture with proprietary hardware, making the elastic multicast service operation challenging.

Software defined networking (SDN) [3] and network function virtualization (NFV) [4] are perceived as two promising paradigms for enabling future softwarized and virtualized networks. Specifically, SDN allows centralized network control by separating the control plane and the data plane, supporting the direct management of flow routing [3], [5], [6]. NFV decouples network functions from dedicated hardware and implements them as virtual network functions (VNFs), which can be flexibly deployed on general commodity servers [4], [7]. In an SDN/NFV-enabled network, a certain service is represented by an ordered chain of VNFs, called a service function chain (SFC) [8], [9], and the corresponding data flow is steered through appropriate locations where VNFs are dynamically placed to accomplish the service. Such a framework reduces capital and operational costs and makes multicast service provisioning and management more flexible and efficient. Nevertheless, benefits from SDN and NFV are accompanied by a series of resource allocation challenges [10], [11]. Among them, one of the main challenges is optimally implementing the SFC orchestration, i.e., orchestrating the VNF placement and routing scheme of SFC to match multicast service and network resources. A typical type of multicast SFC orchestration (MSO) is **Multicast SFC Embedding (MSE)** [12], [13], [14], [15], [16], which specifies the orchestration scheme of the multicast SFC for initial service provisioning. Compared with SFC orchestration for unicast, MSO is more complicated and challenging due to the following features of multicast service.

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The first one is multicast routing. MSO is coupled with multicast routing, as each routing path of source-destination pairs should pass all required VNFs in the SFC. However, jointly optimizing MSO and multicast routing is tricky, especially when simultaneously considering limited network resources and user quality of service (QoS) requirements (e.g., delay requirement). Constructing optimal multicast routing involves the Steiner tree problem [17], which is NP-hard, and it becomes more difficult when respecting the VNF order in the SFC and considering network resource capacity and user QoS requirements [18]. Few works about MSO have well coordinated the above factors. Several works [12], [13] study MSO with the assumption that replication points in multicast routing can only occur after the data flow has been processed by all VNFs in the SFC. Although some works [14], [15] optimize MSO by considering the flexible replication that replication points in multicast routing can exist before and after the flow processing of arbitrary VNFs, either resource capacity (e.g., link bandwidth capacity) or the delay requirement is ignored.

Another universal feature of multicast service is frequent user arrival and departure, i.e., user fluidity. At any time of a multicast service period, some existing users may depart from the service, or some new users may try to access the service. For example, in a live streaming service, audiences may quit or join anytime. The departure of users will release some VNFs or routings and hence change the overall orchestration scheme of the SFC, while newly coming users also require SFC orchestration for service access. Therefore, to accommodate such fluidity of users, in addition to MSE, the following two types of MSO should be taken into account. (i) **Multicast SFC readjustment (MSR)**: adjusting the overall SFC orchestration scheme of all existing users for higher resource utilization as the current scheme may not be optimal after the departure of some users. (ii) **Multicast SFC expanding (MSEP)**: specifying the SFC orchestration schemes for new users meanwhile adjusting existing users' schemes to pursue overall optimality or admit more new users. Here note that MSR and MSEP cannot be simply replaced by reimplementing MSE for all users because multiple additional conditions should be considered during the MSR and MSEP, such as VNF migration and existing user preservation (more details will be explained in Section III-B). Unfortunately, existing MSO works focus heavily on MSE for initial service provisioning. Very little attention is paid to the user fluidity of multicast service, which requires different types of MSO.

Consequently, considering the above two features of multicast service, and in order to fill the gaps in existing MSO works, in this paper, we study the MSO in SDN/NFV-enabled networks encompassing MSE, MSR, and MSEP. In each type of MSO, we jointly consider three key optimization factors: flexible replication in multicast routing, capacitated network resources, and source-destination delay constraint. On the one hand, we study MSO by simultaneously taking the above three factors into consideration, as few existing MSO works have well coordinated them in both model and solution. On the other hand, aside from MSE for initial service provisioning, we also investigate two new types of MSO: MSR and MSEP, to maintain high service performance and resource utilization under the fluidity of users. Specifically, our contributions in this paper are as follows:

1) We define the three types of MSO problems which are MSE, MSR, and MSEP problems, and formulate them with three integer linear programmings (ILPs), aiming at maximizing the network's net revenue. We also prove that all the above MSO problems are NP-hard. To the best of our knowledge, our work is the first to propose and study the MSR and MSEP problems.
2) We jointly consider the three key optimization factors in both model and solution of each MSO problem, which are flexible replication in multicast routing, practical constraints on network resource limitation (including the capacity constraints of node computing resource and link bandwidth resource), and source-destination delay constraint.
3) We develop a novel algorithm source-destination embedding (SDE) and devise three heuristic algorithms called SDE-MSE, SDE-MSR, and SDE-MSEP based on SDE to tackle the three types of MSO problems respectively.
4) We perform extensive simulations to evaluate the proposed three algorithms. Simulation results validate that our algorithms outperform benchmark algorithms in terms of net revenue and acceptance ratio and achieve near-optimal performance.

The remainder of the paper is organized as follows. In Section II, we give an overview of related works. In Section III, we present the system model and define the three types of MSO problems. We mathematically formulate the three MSO problems by three ILPs in Section IV and devise three algorithms to solve them in Section V. In Section VI, we present simulation results to evaluate the performance of the proposed algorithms. Lastly, in Section VII, we conclude this paper.

II. Related Works

The SFC orchestration in SDN/NFV networks for unicast service has been well studied over the past years. The reader is referred to [19], [20] for surveys. For a multicast service case, simply applying a unicast-based SFC orchestration approach to each source-destination pair independently will result in the waste of network resources and high service operation costs [15]. There are a few works about MSO, which mainly focus on MSE for initial service provisioning.

Several works explore MSE on the assumption that multicast replication points can only occur after the flow processing of all VNFs in the SFC [12], [13], [21], [22], [23], [24]. In [21] and its extended work [22], the authors study the SDN/NFV-enabled multicasting problem and propose an online approximation algorithm to solve it. However, it is assumed in their model that all VNFs in the SFC are hosted on one server. Authors in [23] investigate the multicast-aware service function tree embedding problem and propose an algorithm with the objective of minimizing bandwidth consumption. Works [12], [13], [24] explore MSO by considering the delay requirement of multicast service. In [12], the authors

TABLE I
COMPARISON OF EXISTING MSO WORKS

References	**Optimization Factor**				**Involved MSO Type**
	Flexible Replication	Resource Capacity Constraint: Node Computing Capacity Constraint	Resource Capacity Constraint: Link Bandwidth Capacity Constraint	Source-Destination Delay Constraint	
[21] [22] [23]	×	✓	✓	×	MSE
[12] [13]	×	✓	×	✓	MSE
[24]	×	✓	✓	✓	MSE
[25]	✓	×	×	×	MSE
[14]	✓	✓	×	×	MSE
[15] [18] [26] [27] [28] [29] [30]	✓	✓	✓	×	MSE
[31]	✓	✓	×	✓	MSE
[16] [32]	✓	✓	✓	✓	MSE
Our Work	✓	✓	✓	✓	MSE, MSR, MSEP

study the joint optimization problem of VNF placement and routing selection with the consideration of delay constraint. Authors in [13] investigate the delay-aware network slicing problem for multicast service in edge networks. They propose an approximation algorithm and a heuristic algorithm to solve the problem. Nevertheless, link bandwidth capacity is ignored in both [12] and [13]. As a matter of fact, the bandwidth overload of the link will incur severe queuing delay and even congestion. Therefore, the link bandwidth capacity must be respected during the optimization of delay or routing, especially for multicast service which is usually bandwidth-hungry. The reliability and delay-aware MSE problem is discussed in [24].

The restriction on multicast replication points will decrease the flexibility of MSO and further lead to the reduction of feasible solution space. Hence, some works explore MSE with flexible replication in multicast routing [14], [15], [16], [18], [25], [26], [27], [28], [29], [30], [31], [32]. Authors in [25] present approaches for building the SDN/NFV-enabled multicast topology, assuming that an SFC only has one VNF. In [14], the authors investigate the multicast service function tree embedding problem and propose a two-stage heuristic algorithm to tackle it. However, the authors neglect the link bandwidth capacity and delay constraint. Since the delay has a significant impact on user satisfaction [33], the delay constraint should be carefully considered in MSO. Authors in [15] present a multicast service orchestration framework under SDN/NFV-enabled architecture and develop a heuristic algorithm to maximize throughput and minimize overall provisioning cost. In [26], the authors investigate the typical MSE problem, including VNF placement and traffic routing, and propose heuristic algorithms to accommodate single-path and multipath routing cases. The embedding problem of the NFV-enabled multicast SFC is investigated in [27] and [28], while the orchestration problem of multicast SFC in mobile edge cloud networks is explored in [29] and [31]. Moreover, several works explore MSO by jointly considering the flexible replication and the delay requirement of multicast service. In [18], the authors study the multicast-oriented SFC embedding problem and devise a two-stage approach to tackle it. The authors optimize the end-to-end delay by incorporating it into the objective function instead of taking it as a constraint. Authors in [32] discuss the NFV-enabled multicast problem over hybrid infrastructure with consideration of delay constraint. They design a multi-stage heuristic algorithm for solving the problem. The delay-aware multi-source NFV multicast resource optimization problem is studied in [16] and [30]. Nevertheless, the delay constraint considered in [30] is not from each source-destination pair's perspective.

In Table I, we compare our work with existing MSO works. The MSE problems considered in [16] and [32] are the most relevant to our MSE problem. However, in [32], the devised algorithm ignores the link bandwidth capacity and delay constraints of the problem. In [16], the proposed heuristic algorithms assume that the data flow of a multicast request passes through each transmission link at most once, which reduces feasible solution space.

Existing MSO works mainly focus on the MSE problem, and their proposed models or solutions still have the aforementioned gaps in joint optimization of MSO and multicast routing, such as the restriction on replication points and neglect of network resource limitation and delay constraint. On the other hand, to better adapt to the user fluidity of multicast service, it is necessary to implement MSR and MSEP. MSE targets initial service provisioning, and existing models and solutions proposed for MSE are not applicable to MSR and MSEP because MSR and MSEP involve VNF migration and existing user preservation. To address the research gaps, in this paper, we investigate MSO, including MSE, MSR, and MSEP, by carefully considering flexible replication, resource capacity, and source-destination delay constraint in both model and solution.

III. SYSTEM MODEL AND PROBLEM DEFINITION

In this section, we introduce the system model and formally define the three types of MSO problems, namely MSE, MSR, and MSEP problems. The important notations in this paper are listed in Table II.

TABLE II
IMPORTANT NOTATIONS

Notation	Description
Physical Network	
$G=(V,E)$	physical network G with node set V and link set E
V_{for}	the set of forwarding nodes
V_{ser}	the set of service nodes
$u,v\in V$	a node u and a node v
c_v	the residual computing capacity of $v\in V$
N_v	the set of v's $(v\in V)$ neighbor nodes
$e\in E$	a link e
b_e	the residual bandwidth capacity of $e\in E$
l_e	the delay of $e\in E$
$\hat{E}$	the set of directed links
$e_{u,v}\in\hat{E}$	a directed link from $u\in V$ to $v\in N_u$
MSO Request	
r	the MSO request
r_{MSE}	the MSE request
r_{MSR}	the MSR request
r_{MSEP}	the MSEP request
s	the source node
D	the set of destination nodes
D_{eu}	the set of existing user destination nodes
D_{nu}	the set of new user destination nodes
$d\in D$	a destination node d
F	the set of ordered VNFs
$\hat{F}$	the redenoted set of ordered VNFs
$f_i\in\hat{F}$	a VNF f_i $(i=0,1,...,\|F\|+1)$
ω_{f_i}	the required computing resource of $f_i\in\hat{F}$
H	the set of virtual links
$h_j\in H$	a virtual link h_j $(j=1,2,...,\|F\|+1)$
T	the set of virtual link delay constraints
$t_j\in T$	the virtual link delay constraint of $h_j\in H$
b	the bandwidth requirement
Γ	the overall SFC orchestration scheme of all existing users before MSO implementation
τ_v^{Γ}	the consumed computing resource at $v\in V$ by Γ
$\varepsilon_{uv}^{\Gamma}$	the consumed bandwidth resource at link consisting of $u\in V$ and $v\in V$ by Γ
$\varphi_{d,u}^{\Gamma,f_i}$	indicates whether $f_i\in\hat{F}$ at $u\in V$ processes the flow entering the $d\in D_{eu}$ in Γ
κ	the revenue of successfully admitting an new user
Decision Variables	
δ_d	indicates whether new user at the $d\in D_{nu}$ is admitted after MSO implementation
$x_{d,v}^{f_i}$	indicates whether $v\in V$ is used to place $f_i\in\hat{F}$ processing the flow entering the $d\in D$ after MSO implementation
$y_{d,e_{u,v}}^{h_j}$	indicates whether $e_{u,v}\in\hat{E}$ is used to deploy the $h_j\in H$ for $d\in D$ after MSO implementation
$z_v^{f_i}$	indicates whether $v\in V$ is used to place $f_i\in\hat{F}$ after MSO implementation
$w_{e_{u,v}}^{h_j}$	indicates whether $e_{u,v}\in\hat{E}$ is used to deploy the $h_j\in H$ after MSO implementation
Cost Factors	
α	the cost of unit node computing resource
β	the cost of unit link bandwidth resource
$\gamma_{u,v}^{f_i}$	the cost of migrating an instance of $f_i\in\hat{F}$ from $u\in V$ to $v\in V$

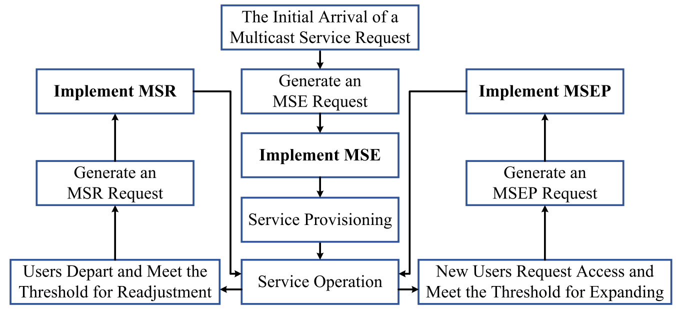


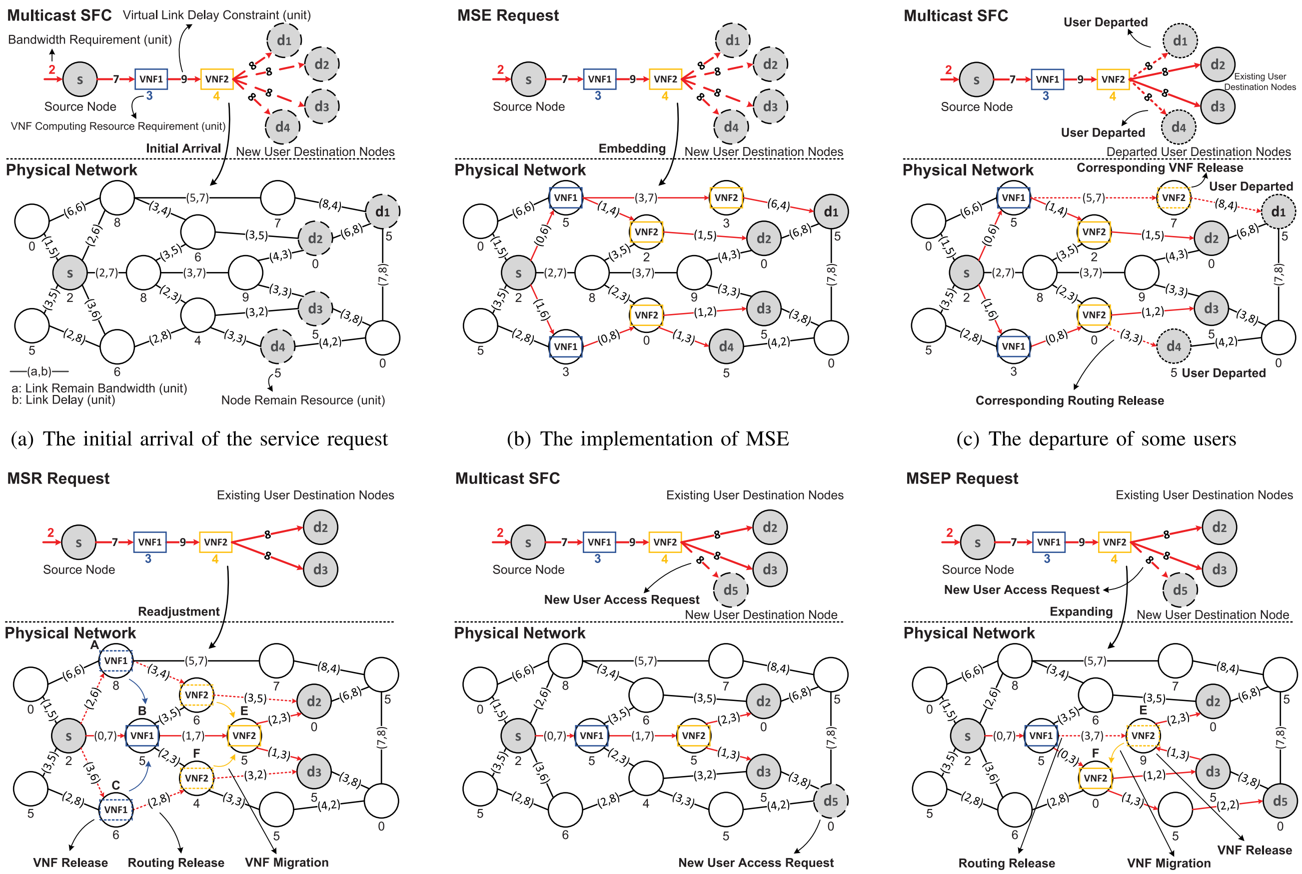
Fig. 1. The Overall MSO Process.

A. System Model

1) Overall MSO Process: We consider the overall MSO process as depicted in Fig. 1. As shown in the figure, the initial arrival of a multicast service request will generate an MSE request, and the orchestrator implements MSE to provision the service. During the service, when some users depart from the service and the threshold for readjustment is met, an MSR request is generated and the orchestrator implements MSR to adjust the orchestration of the multicast SFC. Similarly, when some new users request access during the service and the threshold for expanding is reached, an MSEP request is generated and the orchestrator implements MSEP for user expanding. In this paper, we focus on the implementations of MSE, MSR, and MSEP, and how to define the thresholds for readjustment and expanding is beyond the scope of this paper and we will pursue it in the future. Fig. 2 presents the examples of the three MSO implementations.

2) Physical Network: We consider a capacitated physical network $G = (V = V_{for} \cup V_{ser}, E)$, where V, V_{for}, V_{ser}, and E stand for the sets of nodes, forwarding nodes, service nodes, and links, respectively. Each node $v \in V$ is either a forwarding node or a service node, and here we use c_v to represent its residual computing capacity. A forwarding node (e.g., switch) is only in charge of forwarding traffic to neighbor nodes (i.e., $c_v = 0$ when $v \in V_{for}$), while a service node (e.g., server) is capable of both forwarding traffic and operating VNFs. Besides, we use N_v to denote the set of node v's $(v \in V)$ neighbor nodes. Each link $e \in E$ has a residual bandwidth capacity b_e and an associated delay l_e. To better describe the flow direction, we use $e_{u,v} \in \hat{E}$ $(u \in V, v \in N_u)$ to represent the directed link from node u to node v and denote the set of all directed links as $\hat{E}$ (here we have $|\hat{E}| = 2|E|$).

3) MSO Request: We denote an MSO request as an eight tuple $r = (s, D, F, H, T, b, \Gamma, \kappa)$. In particular, $s \in V$ is the source node, and $D \subset V$ is the set of user destination nodes (we assume $s \notin D$). D consists of two kinds of user destination node sets D_{eu} and D_{nu}, i.e., $D = D_{eu} \cup D_{nu}$. D_{eu} is the destination node set for the existing users who are being served, while D_{nu} is the destination node set for the new users who try to access the service. The MSO request is either an MSE request r_{MSE} $(D_{eu} = \emptyset$, $D_{nu} \neq \emptyset)$ or an MSR request r_{MSR} $(D_{eu} \neq \emptyset$, $D_{nu} = \emptyset)$, or an MSEP request r_{MSEP} $(D_{eu} \neq \emptyset$, $D_{nu} \neq \emptyset)$, according to the constitution of D. $F = (f_1, f_2, \ldots, f_{|F|})$ is the set of ordered VNFs, namely the SFC, where f_g is the g-th VNF in F $(g = 1, 2, \ldots, |F|)$. For a better description in later formulation, we regard s and $\forall d \in D$ as dummy VNFs and set $s = f_0$ and $d = f_{|F|+1}$. Then the SFC is re-denoted as $\hat{F} = (f_0, f_1, \ldots, f_{|F|}, f_{|F|+1})$.



(a) The initial arrival of the service request (b) The implementation of MSE (c) The departure of some users

(d) The implementation of MSR (e) The access request of the new user (f) The implementation of MSEP

Fig. 2. The examples of MSO implementations including MSE, MSR, and MSEP.

We regard $|\hat{F}|$ as the length of the SFC, and each VNF $f_i \in \hat{F}$ ($i = 0, 1, \ldots, |F| + 1$) requires ω_{f_i} ($\omega_{f_0} = \omega_{f_{|F|+1}} = 0$) computing resource to be instantiated at the service node. $H = (h_1, h_2, \ldots, h_{|F|+1})$ is the set of virtual links, and each virtual link h_j ($j = 1, 2, \ldots, |F| + 1$) represents the logical link connecting two adjacent VNFs f_{j-1} and f_j in $\hat{F}$. Compared with end-to-end delay constraint, we consider a more strict delay constraint mode that there is a delay guarantee between two adjacent VNFs in SFC, i.e., virtual link delay constraint [34]. $T = (t_1, t_2, \ldots, t_{|F|+1})$ is the set of virtual link delay constraints, where t_j is the delay constraint of h_j. b is the bandwidth requirement of r. Γ is the overall SFC orchestration scheme of all existing users of multicast service before MSO implementation ($\Gamma = Null$ when $r = r_{MSE}$). Here we assume that all the information about Γ is known. Finally, κ is the revenue of successfully admitting a new user for r ($\kappa = 0$ when $r = r_{MSR}$).

B. Problem Definition

1) Multicast SFC Embedding (MSE) Problem: The role of MSE is initial service provisioning for the initially arrived multicast service request. Fig. 2(a) shows an example of the initial arrival of a multicast service request. In most scenarios, the demand tends to exceed the supply [35]. Therefore, considering the limited network resources and delay constraint, there is no guarantee that all new users' access requests can be satisfied. We assume that each successfully admitted new user could bring certain revenue. The objective of MSE is to maximize net revenue by admitting more new users and reducing resource costs. Here we define the MSE problem as follows.

Definition 1 (MSE Problem): Given $G = (V, E)$ and MSE request r_{MSE}, under the resource capacity and delay constraints, the MSE problem is to find an overall VNF placement and multicast routing scheme for all new users to maximize the net revenue that the new user admission revenue minus the resource costs.

Fig. 2(b) presents an example of implementing MSE. In this example, all the new users at $d_1 \sim d_4$ have been admitted. When we assume that the admission of a new user brings 10 monetary units and both per unit node computing and link bandwidth resource consumption cost 1 monetary unit, the net revenue in this example is $10 \times 4 - (3 \times 2 + 4 \times 3) - 2 \times 9 = 22$ monetary units.

2) Multicast SFC Readjustment (MSR) Problem: When some of users depart from the service, the corresponding VNFs and routings will be released. Fig. 2(c) shows an example of user departure and the release of corresponding VNFs and routings. Such release will change the overall orchestration scheme of the multicast SFC and may make it not optimal. In this context, we can implement MSR to adjust the existing users' orchestration schemes for higher resource utilization. Nevertheless, the following conditions should be carefully considered in MSR. (i) The adjustment in MSR involves the migration of VNFs, which incurs the operational cost [36]. (ii) The VNF migration during the service can lead to service jitter or short suspension, and some high-demanding multicast services may be sensitive to them. (iii) The existing users cannot be rejected during the MSR implementation. The above conditions make MSR significantly different from MSE, and we cannot simply reimplement MSE for all existing users to replace MSR. In order to consider the first two conditions in an integrated manner, we treat the negative effects (i.e., the above jitter or short suspension) brought by VNF migration as a kind of penalty cost because such effects will reduce the revenue of some high-demanding services. The penalty cost can be modeled by investigating historic marketing statistics. Then we combine the operational cost and the penalty cost as VNF migration cost. In MSR, we regard the saved resource costs through adjustment as revenue, and the objective of MSR is to maximize net revenue by saving more resource costs and reducing the VNF migration cost. Now we define the MSR problem as follows.

Definition 2 (MSR Problem): Given $G = (V, E)$ and MSR request r_{MSR}, under the resource capacity and delay constraints, the MSR problem is to adjust the overall VNF placement and multicast routing scheme for all existing users to maximize the net revenue that the resource-saving revenue minus the VNF migration cost.

Fig. 2(d) presents an example of implementing MSR. In this example, the previous overall SFC orchestration scheme (denoted by the dotted line) has been adjusted to a better one (denoted by the solid line), saving more node computing and link bandwidth resources. Here note that, for all the existing users in this example, their corresponding VNF instances have been migrated to other nodes. For example, for the existing user at d_3, the corresponding VNF1 instance has been migrated from node C to node B, while the corresponding VNF2 instance has been migrated from node F to node E. When we assume that each VNF migration cost is 1 monetary unit, the net revenue in this example is $(3 \times 2 + 4 \times 2 + 2 \times 6) - (3 + 4 + 2 \times 4) - 1 \times 4 = 7$ monetary units.

3) Multicast SFC Expanding (MSEP) Problem: The role of MSEP is to admit the new users who try to access the service midway. Fig. 2(e) shows an example of the access request of a new user. However, we should simultaneously adjust the SFC orchestration schemes of existing users to pursue overall optimization or admit more new users. The following conditions should be carefully considered in MSEP. (i) Not all new users can be admitted due to the network resource limitation and delay constraint. (ii) The adjustment in MSEP involves the VNF migration, resulting in the VNF migration cost. (iii) Existing users must be preserved during MSEP. Therefore, MSEP is also different from MSE and cannot be replaced by MSE. The objective of MSEP is to maximize the net revenue by admitting more new users, reducing new users' resource costs, saving more existing users' resource costs, and reducing the existing users' VNF migration cost. Here we define the MSEP problem as follows.

Definition 3 (MSEP Problem): Given $G = (V, E)$ and MSEP request r_{MSEP}, under the resource capacity and delay constraints, the MSEP problem is to adjust and expand the overall VNF placement and multicast routing scheme for all existing and new users to maximize the net revenue that the sum of new user admission revenue and existing user resource-saving revenue minus the sum of the new user resource costs and the existing user VNF migration cost.

Fig. 2(f) presents an example of implementing MSEP. In this example, the new user at d_5 has been admitted. Here note that the instance of VNF2 has been migrated from node E to node F to satisfy the last virtual link delay constraint of d_5, which incurs the VNF migration cost from the existing users at d_2 and d_3. The net revenue in this example is $10 \times 1 + ((3 + 4 + 2 \times 4) - (3 + 4 + 2 \times 5)) - 2 \times 2 - 1 \times 2 = 2$ monetary units.

IV. Problem Formulation

In this section, we mathematically formulate the three types of MSO problems defined in previous section.

A. Decision Variables

We first introduce a binary decision variable $\delta_d \in \{0, 1\}$ $(\forall d \in D_{nu})$ that takes the value as 1 if the new user at d is admitted after MSO implementation and 0 otherwise. We then define two binary decision variables from each user destination node's perspective: user node decision variable $x_{d,v}^{f_i}$ and user link decision variable $y_{d,e_{u,v}}^{h_j}$ as follows.

$$x_{d,v}^{f_i} = \begin{cases} 1 & \textit{if node } v \in V \textit{ is used to place VNF } f_i \in \hat{F} \\ & (i = 0, 1, \ldots, |F| + 1) \textit{ processing the flow} \\ & \textit{that enters user destination node } d \in D \\ & \textit{after MSO implementation}, \\ 0 & \textit{otherwise}. \end{cases}$$

$$y_{d,e_{u,v}}^{h_j} = \begin{cases} 1 & \textit{if link } e_{u,v} \in \hat{E} \textit{ is used to deploy virtual link} \\ & h_j \in H (j = 1, 2, \ldots, |F| + 1) \ (\textit{i.e., transfer the} \\ & \textit{flow that has been processed by } f_{j-1} \textit{ but not} \\ & \textit{by } f_j) \textit{ for user destination node } d \in D \\ & \textit{after MSO implementation}, \\ 0 & \textit{otherwise}. \end{cases}$$

We also define two binary decision variables from the overall perspective: overall node decision variable $z_v^{f_i}$ and overall link decision variable $w_{e_{u,v}}^{h_j}$ as follows.

$$z_v^{f_i} = \begin{cases} 1 & \textit{if node } v \in V \textit{ is used to place VNF } f_i \in \hat{F} \\ & (i = 0, 1, \ldots, |F| + 1) \textit{ after MSO implementation}, \\ 0 & \textit{otherwise}. \end{cases}$$

$$w_{e_{u,v}}^{h_j} = \begin{cases} 1 & \textit{if link } e_{u,v} \in \hat{E} \textit{ is used to deploy the virtual} \\ & \textit{link } h_j \in H (j = 1, 2, \ldots, |F| + 1) \textit{ after MSO} \\ & \textit{implementation}, \\ 0 & \textit{otherwise}. \end{cases}$$

B. Constraints

Constraints (1) and (2) state that the multicast flow starts from the source node and ends at the destination nodes.

$$x_{d,s}^{f_0} = x_{d,d}^{f_{|F|+1}} = 1, \quad \forall d \in D_{eu} \tag{1}$$

$$x_{d,s}^{f_0} = x_{d,d}^{f_{|F|+1}} = \delta_d, \quad \forall d \in D_{nu} \tag{2}$$

Constraints (3) and (4) guarantee that the flow entering each existing user or admitted new user destination node is processed by the required VNFs in the SFC only once.

$$\sum_{v \in V} x_{d,v}^{f_i} = 1, \quad \forall f_i \in \hat{F}, d \in D_{eu} \tag{3}$$

$$\sum_{v \in V} x_{d,v}^{f_i} = \delta_d, \quad \forall f_i \in \hat{F}, d \in D_{nu} \tag{4}$$

Constraints (5) and (6) ensure that the flow entering each user destination node cannot split.

$$\sum_{u \in N_v} y_{d,e_{u,v}}^{h_j} \leq 1, \quad \forall h_j \in H, d \in D, v \in V \tag{5}$$

$$\sum_{u \in N_v} y_{d,e_{v,u}}^{h_j} \leq 1, \quad \forall h_j \in H, d \in D, v \in V \tag{6}$$

Constraint (7) guarantees the flow conservation at each node from each user destination node's perspective.

$$\sum_{u \in N_v} y_{d,e_{v,u}}^{h_j} - \sum_{u \in N_v} y_{d,e_{u,v}}^{h_j} = x_{d,v}^{f_{j-1}} - x_{d,v}^{f_j}, \quad \forall h_j \in H, d \in D, v \in V \tag{7}$$

Constraints (8) and (9) express the relationships between user decision variables and overall decision variables. Specifically, constraint (8) indicates that $z_v^{f_i} = 1$ if $x_{d,v}^{f_i} = 1$ for any $d \in D$, as we only need to place one instance of f_i at node v to process one copy of flow in multicast even if multiple user destination nodes are served by f_i placed at node v. Similarly, constraint (9) indicates that $w_{e_{u,v}}^{h_j} = 1$ if $y_{d,e_{u,v}}^{h_j} = 1$ for any $d \in D$, as we only need to transfer one copy of flow in multicast.

$$x_{d,v}^{f_i} \leq z_v^{f_i}, \quad \forall f_i \in \hat{F}, d \in D, v \in V \tag{8}$$

$$y_{d,e_{u,v}}^{h_j} \leq w_{e_{u,v}}^{h_j}, \quad \forall h_j \in H, d \in D, e_{u,v} \in \hat{E} \tag{9}$$

Constraint (10) ensures that the residual computing capacity of each node is not violated, where τ_v^{Γ} denotes the consumed computing resource at node v by scheme Γ (τ_v^{Γ} is a known value and $\tau_v^{\Gamma} = 0$ when $r = r_{MSE}$).

$$\sum_{f_i \in \hat{F}} \omega_{f_i} \cdot z_v^{f_i} \leq c_v + \tau_v^{\Gamma}, \quad \forall v \in V \tag{10}$$

Constraint (11) guarantees that the residual bandwidth capacity of each link is not violated, where $\varepsilon_{uv}^{\Gamma}$ denotes the consumed bandwidth resource at the link consisting of node u and node v by scheme Γ. ($\varepsilon_{uv}^{\Gamma}$ is a known value and $\varepsilon_{uv}^{\Gamma} = 0$ when $r = r_{MSE}$).

$$\sum_{h_j \in H} b \cdot \left(w_{e_{u,v}}^{h_j} + w_{e_{v,u}}^{h_j} \right) \leq b_{e_{u,v}} + \varepsilon_{uv}^{\Gamma}, \quad \forall e_{u,v} \in \hat{E}, u < v \tag{11}$$

Constraint (12) ensures the delay constraint of each virtual link is not violated from each user destination node's perspective.

$$\sum_{e_{u,v} \in \hat{E}} l_{e_{u,v}} \cdot y_{d,e_{u,v}}^{h_j} \leq t_j, \quad \forall h_j \in H, d \in D \tag{12}$$

C. Revenues and Costs

The total new user admission revenue of all new users after MSO implementation is calculated as formula (13).

$$R_{adm} = \sum_{d \in D_{nu}} \kappa \cdot \delta_d \tag{13}$$

The total resource cost of all users after the MSO implementation is expressed as formula (14), where α and β are the costs of unit node computing resource and unit link bandwidth resource, respectively. In (14), the first term is total computing cost, while the second term is total bandwidth cost.

$$C_{res} = \sum_{f_i \in \hat{F}} \sum_{v \in V} \alpha \cdot \omega_{f_i} \cdot z_v^{f_i} + \sum_{h_j \in H} \sum_{e_{u,v} \in \hat{E}} \beta \cdot b \cdot w_{e_{u,v}}^{h_j} \tag{14}$$

The total resource-saving revenue of all users after MSO implementation is calculated as formula (15), where C_{res}^{Γ} is the total resource cost of Γ (C_{res}^{Γ} is a known value).

$$R_{sav} = C_{res}^{\Gamma} - C_{res} \tag{15}$$

The total VNF migration cost of all existing users after MSO implementation is expressed as formula (16). In (16), $\gamma_{u,v}^{f_i}$ is the migration cost coefficient, and it represents the cost of migrating an instance of f_i from node u to node v ($\gamma_{u,v}^{f_i} = 0$ when $u = v$) for r in the network, and $\varphi_{d,u}^{\Gamma,f_i}$ takes the value as 1 if f_i placed at node u processes the flow entering the existing user destination node d in Γ and 0 otherwise ($\varphi_{d,u}^{\Gamma,f_i}$ is a known value).

$$C_{mig} = \sum_{d \in D_{eu}} \sum_{f_i \in \hat{F}} \sum_{u \in V} \sum_{v \in V} \gamma_{u,v}^{f_i} \cdot \varphi_{d,u}^{\Gamma,f_i} \cdot x_{d,v}^{f_i} \tag{16}$$

D. MSO Problems

Now we formulate the three types of MSO problems as the following three integer linear programmings (ILPs) aiming to maximize the net revenues.

MSE Problem ($r = r_{MSE}$):

$$Maximize : R_{adm} - C_{res} \tag{17a}$$

$$Subject\ to : (2), (4), \text{ and } (5) - (12) \tag{17b}$$

MSR Problem ($r = r_{MSR}$):

$$Maximize : R_{sav} - C_{mig} \tag{18a}$$

$$Subject\ to : (1), (3), \text{ and } (5) - (12) \tag{18b}$$

MSEP Problem ($r = r_{MSEP}$):

$$Maximize : R_{adm} + R_{sav} - C_{mig} \tag{19a}$$

$$Subject\ to : (1) - (12) \tag{19b}$$

Theorem: All the above three formulated MSO problems are NP-hard.

Proof: We prove the theorem by a reduction from the Steiner tree problem [17] in polynomial time. Assume an MSO request r with multiple user destination nodes has only one VNF function f (i.e., $|F| = 1$), and each virtual link of r has fully loose delay constraint. We consider a physical network G, and each link in G has a fully sufficient bandwidth resource. The source node of r is a leaf node in G and is directly connected to only one service node in G (this service node has a fully sufficient computing resource). In this context, regardless of the MSO type, the objective has become to find an overall VNF placement and multicast routing scheme for all users which can achieve the minimum total resource cost, as all new users will be admitted and the VNF cannot be migrated in this scenario. The optimal solution can be obtained by following two steps: (i) place one instance of f at the service node and establish the routing from the source node to the service node; (ii) construct the Steiner tree routing with minimum total hop counts from service node to all the user destination nodes. Although the first step is performed in polynomial time, the second step is NP-hard. Therefore, all the three types of MSO problems are NP-hard. ■

A general way to solve an ILP with NP-hardness is by applying the branch-and-bound (B&B) [37] algorithm (B&B is also the main framework of current ILP solvers.). Although B&B can obtain the optimal solution, the time complexity is unacceptable when the problem scale becomes larger. Some techniques like Dantzig-Wolfe decomposition (DWD) [38] and column generation (CG) [39] can be used to accelerate the solving process by coupling them with B&B to form the branch-and-price (B&P) [40] algorithm. However, it is hard to use B&P to solve our MSO problems. The B&P algorithm requires the constraints of the problem to match a block angular structure [41] so that the problem can be decomposed by multiple similar subproblems. In our MSO problems, the existence of admission control constraints (constraints (1) and (2)) and variable relation constraints (constraints (8) and (9)) makes the overall constraints in our MSO problems hard to be characterized by the block angular structure. Besides, B&P still takes much time to obtain the solution when the problem scale is large. Authors in [14] propose a two-stage MSE algorithm that combines the dynamic programming [42] and the Steiner tree approximation algorithm [43]. As the Steiner tree approximation algorithm can achieve an approximation ratio as small as 1.39 [43], their algorithm can also guarantee an approximation ratio that is smaller than a determinate value. Nevertheless, in our case, as we jointly consider the flexible replication, link bandwidth capacity, and virtual link delay constraint, the Steiner tree approximation algorithm is also not applicable to our MSO problems. On the one hand, the topology structure may change along with the bandwidth allocation (in our model, one link is regarded as disconnected when its bandwidth is insufficient for the data flow), making it difficult to use the Steiner tree approximation algorithm, especially when considering the flexible replication. On the other hand, the constructed approximate Steiner tree with small bandwidth consumption may not satisfy the virtual delay constraint.

Therefore, in this paper, we devise three polynomial-time heuristic algorithms to respectively solve the three NP-hard MSO problems which are introduced in detail in the subsequent section.

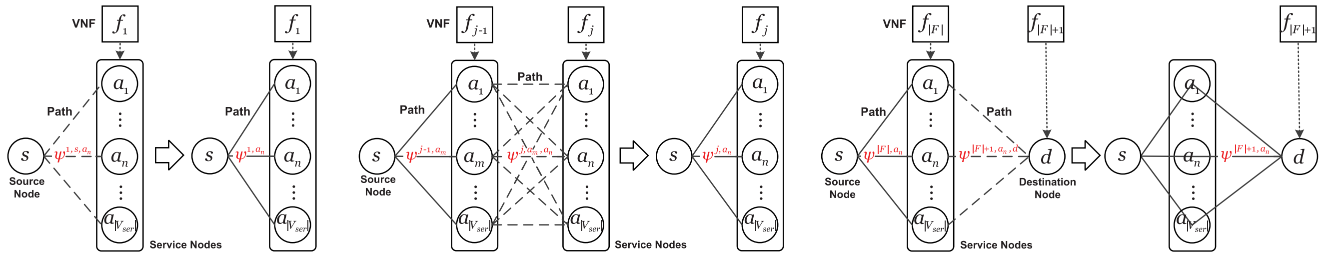
V. Heuristic Approaches

In this section, we first propose the source-destination embedding (SDE) algorithm and then introduce three heuristic algorithms, SDE-MSE, SDE-MSR, and SDE-MSEP, designed based on SDE, to respectively solve the formulated MSE, MSR, and MSEP problems.

A. Source-Destination Embedding (SDE)

The role of SDE is to place VNFs and specify routing for one source-destination pair. SDE relies on four kinds of record matrices: VNF record matrix $\mathbf{P} = \{p_{jv}\}_{(|F|+1)\times|V|}$, virtual link record matrix $\mathbf{Q}^{h_j} = \{q_{uv}^{h_j}\}_{|V|\times|V|}$, residual computing capacity record vector $\mathbf{c} = (c_{a_1}, c_{a_2}, \ldots, c_{a_{|V_{ser}|}})$, and residual bandwidth capacity record symmetric matrix $\mathbf{B} = \{b_{uv}\}_{|V|\times|V|}$. In $\mathbf{P}$, element $p_{jv} \in \{0,1\}$ is recorded to 1 if $f_j \in \hat{F}$ $(j = 1, 2, \ldots, |F|+1)$ of any source-destination pair has been placed at node $v \in V$. Element $q_{uv}^{h_j} \in \{0,1\}$ in $\mathbf{Q}^{h_j}$ is recorded to 1 if $e_{u,v}$ has been specified to deploy virtual link $h_j \in H$ $(j = 1, 2, \ldots, |F|+1)$ for any source-destination pair. In $\mathbf{c}$, element c_{a_n} is the current residual computing capacity of service node $a_n \in V_{ser}$ $(n = 1, 2, \ldots, |V_{ser}|)$. In $\mathbf{B}$, element b_{uv} is the current residual bandwidth capacity of the link consisting of node u and node v. Here note that if $e_{u,v}$ does not exist, the corresponding elements $q_{uv}^{h_j}$ and b_{uv} are equal to 0.

Now we explain SDE in detail. SDE decides VNF placement and routing for one source-destination pair by constantly establishing the stage graph, as shown in Fig. 3. The circles in the figure represent nodes, including source, destination, and service nodes, while the line connecting two circles denotes the path. There are at most $|F| + 1$ stage graphs that need to be established, and the j-th stage $(j = 1, \ldots, |F| + 1)$ deals with the placement of f_j and the deployment of h_j. Each service node $a_n \in V_{ser}$ $(n = 1, 2, \ldots, |V_{ser}|)$ is associated with a record in each stage: a five tuple $\sigma^{j,a_n} = (\mathbf{c}^{j,a_n}, \mathbf{B}^{j,a_n}, \eta^{j,a_n}, \psi^{j,a_n}, \xi^{j,a_n})$. Each element in σ^{j,a_n} records the assumed information from a_n's perspective after establishing the graph of the j-th stage. Specifically, $\mathbf{c}^{j,a_n} = (c_{a_1}^{j,a_n}, c_{a_2}^{j,a_n}, \ldots, c_{a_{|V_{ser}|}}^{j,a_n})$ is the assumed residual computing capacity vector, where element $c_{a_m}^{j,a_n}$ is the assumed residual computing capacity of service node a_m $(m = 1, 2, \ldots, |V_{ser}|)$. $\mathbf{B}^{j,a_n} = \{b_{uv}^{j,a_n}\}_{|V|\times|V|}$ is the assumed residual bandwidth capacity symmetric matrix, where element b_{uv}^{j,a_n} is the assumed residual bandwidth capacity of the link consisting of node u and node v. η^{j,a_n} is a j tuple, and it is the assumed placement strategy of $f_1, f_2, \ldots, f_j$ whose element is the assumed node placing the corresponding VNF. ψ^{j,a_n} is the assumed routing path from the source node to the node placing f_j, represented by the solid line connecting the two corresponding nodes in Fig. 3. ξ^{j,a_n} is the assumed accumulative cost. Additionally, source node s also has record $\sigma^{0,s} = (\mathbf{c}^{0,s}, \mathbf{B}^{0,s}, \eta^{0,s}, \psi^{0,s}, \xi^{0,s})$ and



(a) Establish the first stage graph (b) Establish the j-th $(2 \le j \le |F|)$ stage graph (c) Establish the last stage graph

Fig. 3. Establish stage graphs.

here $\mathbf{c}^{0,s} = \mathbf{c}$, $\mathbf{B}^{0,s} = \mathbf{B}$, $\eta^{0,s} = Null$, $\psi^{0,s} = Null$, and $\xi^{0,s} = 0$.

The process of establishing each stage graph is actually obtaining the records of all service nodes through calculations. SDE gets the record of $a_n \in V_{ser}$ in the j-th stage, namely σ^{j,a_n}, by the following steps. **First**, SDE assumes that f_j is placed at node a_n (or d when $j = |F| + 1$). **Second**, SDE calculates ξ^{j,a_n} according to

$$\xi^{j,a_n} = \begin{cases} \lambda^{1,s,a_n} & j = 1, \\ \min_{a_m \in V_{ser}} \left\{ \xi^{j-1,a_m} + \lambda^{j,a_m,a_n} \right\} & 2 \le j \le |F|, \\ \xi^{|F|,a_n} + \lambda^{|F|+1,a_n,d} & j = |F| + 1. \end{cases} \tag{20}$$

In equation (20), λ^{1,s,a_n}, λ^{j,a_m,a_n}, and $\lambda^{|F|+1,a_n,d}$ are assumed stage costs and they are respectively calculated by

$$\lambda^{1,s,a_n} = \lambda^{1,s,a_n}_{com} + \lambda^{1,s,a_n}_{ban} + \lambda^{1,a_n}_{mig}, \tag{21}$$

$$\lambda^{j,a_m,a_n} = \lambda^{j,a_m,a_n}_{com} + \lambda^{j,a_m,a_n}_{ban} + \lambda^{j,a_n}_{mig}, \tag{22}$$

$$\lambda^{|F|+1,a_n,d} = \lambda^{|F|+1,a_n,d}_{ban}. \tag{23}$$

In equations (21)-(23), λ^{j,a',a_n}_{com} $(a' \in \{s\} \cup V_{ser})$, $\lambda^{j,a',a''}_{ban}$ $(a'' \in \{d\} \cup V_{ser})$, and λ^{j,a_n}_{mig} are assumed computing cost, bandwidth cost, and VNF migration cost, respectively. λ^{j,a',a_n}_{com} is the cost of the consumed computing resource by f_j when assuming $f_1, f_2, \ldots, f_{j-1}$ are placed according to $\eta^{j-1,a'}$ (note that f_{j-1} is placed at node a_m in η^{j-1,a_m} when $2 \le j \le |F| + 1$) and f_j is placed at node a_n, and it is calculated by

$$\lambda^{j,a',a_n}_{com} = \begin{cases} 0 & p_{j\,a_n} = 1, \\ \alpha \cdot \omega_{f_j} & p_{j\,a_n} = 0 \ and \ c^{j-1,a'}_{a_n} \ge \omega_{f_j}, \\ \infty & p_{j\,a_n} = 0 \ and \ c^{j-1,a'}_{a_n} < \omega_{f_j}. \end{cases} \tag{24}$$

Equation (24) indicates that we do not need to place f_j again when f_j has been placed at node a_n by other source-destination pairs. Otherwise, we only place f_j at node a_n when residual computing capacity is enough. $\lambda^{j,a',a''}_{ban}$ is the bandwidth cost of assumed routing path $\psi^{j,a',a''}$ from node a' to node a'' which is represented by the dash line connecting the two corresponding nodes in Fig. 3. $\lambda^{j,a',a''}_{ban}$ and $\psi^{j,a',a''}$ are obtained by constructing link bandwidth cost adjacency matrix $\mathbf{O}^{j,a'} = \{o^{j,a'}_{uv}\}_{|V|\times|V|}$ and link delay adjacency matrix $\mathbf{L}^{j,a'} = \{l^{j,a'}_{uv}\}_{|V|\times|V|}$, whose elements are respectively calculated by

$$o^{j,a'}_{uv} = \begin{cases} 0 & u = v \ or \ q^{h_j}_{uv} = 1, \\ \beta \cdot b & u \ne v \ and \ q^{h_j}_{uv} = 0 \ and \ b^{j-1,a'}_{uv} \ge b, \\ \infty & u \ne v \ and \ q^{h_j}_{uv} = 0 \ and \ b^{j-1,a'}_{uv} < b. \end{cases} \tag{25}$$

$$l^{j,a'}_{uv} = \begin{cases} 0 & u = v, \\ l_{e_{u,v}} & u \ne v \ and \ \left(q^{h_j}_{uv} = 1 \ or \ b^{j-1,a'}_{uv} \ge b\right), \\ \infty & u \ne v \ and \ q^{h_j}_{uv} = 0 \ and \ b^{j-1,a'}_{uv} < b. \end{cases} \tag{26}$$

Equation (25) and equation (26) indicate that we only need to transfer one copy of the flow if link $e_{u,v}$ has been used to deploy virtual link h_j by other source-destination pairs. Otherwise, we can only use the links with enough residual bandwidth capacity to deploy h_j. It is worth mentioning that although we can obtain the routing path from node a' to node a'' with minimum bandwidth cost by finding the shortest path (SP) in $\mathbf{O}^{j,a'}$, there is no guarantee that this path satisfies virtual link delay constraint t_j. Therefore, SDE finds the K shortest paths (KSP) $\psi^{j,a',a''}_1, \psi^{j,a',a''}_2, \ldots, \psi^{j,a',a''}_K$ (with corresponding bandwidth costs $\mu^{j,a',a''}_1, \mu^{j,a',a''}_2, \ldots, \mu^{j,a',a''}_K$ respectively) from node a' to node a'' in $\mathbf{O}^{j,a'}$ and successively checks whether they satisfy t_j. If the k-th shortest path $\psi^{j,a',a''}_k$ $(k = 1, 2, \ldots, K)$ satisfies t_j, there are $\lambda^{j,a',a''}_{ban} = \mu^{j,a',a''}_k$ and $\psi^{j,a',a''} = \psi^{j,a',a''}_k$. If none of them satisfies t_j, SDE then finds the shortest path $\psi^{j,a',a''}_L$ (with corresponding bandwidth cost $\mu^{j,a',a''}_L$ calculated through $\mathbf{O}^{j,a'}$) from node a' to node a'' in $\mathbf{L}^{j,a'}$. If $\psi^{j,a',a''}_L$ satisfies t_j, there are $\lambda^{j,a',a''}_{ban} = \mu^{j,a',a''}_L$ and $\psi^{j,a',a''} = \psi^{j,a',a''}_L$. Otherwise, $\lambda^{j,a',a''}_{ban} = \infty$ and $\psi^{j,a',a''} = Null$. The procedures of obtaining $\lambda^{j,a',a''}_{ban}$ and $\psi^{j,a',a''}$ are depicted in Algorithm 1. λ^{j,a_n}_{mig} is the VNF migration cost when assuming f_j is placed at node a_n, and it is calculated by

$$\lambda^{j,a_n}_{mig} = \begin{cases} \sum_{u \in V} \gamma^{f_j}_{u,a_n} \cdot \varphi^{\Gamma,f_j}_{d,u} & d \in D_{eu}, \\ 0 & d \in D_{nu}. \end{cases} \tag{27}$$

Third, SDE constructs η^{j,a_n} and ψ^{j,a_n} through node a^*, which is found according to

$$a^* = \begin{cases} s & j = 1, \\ \arg\min_{a_m \in V_{ser}} \left\{ \xi^{j-1,a_m} + \lambda^{j,a_m,a_n} \right\} & 2 \le j \le |F|, \\ a_n & j = |F| + 1. \end{cases} \tag{28}$$

Algorithm 1: Calculate the Assumed Bandwidth Cost and Assumed Routing Path

Input: $a' \in s \cup V_{ser}$, $a'' \in d \cup V_{ser}$, $\mathbf{O}^{j,a'}$, $\mathbf{L}^{j,a'}$, t_j.
Output: Assumed Bandwidth Cost $\lambda_{ban}^{j,a',a''}$, assumed routing path $\psi^{j,a',a''}$ from a' to a''.

$(\psi_1^{j,a',a''}, \ldots, \psi_K^{j,a',a''}; \mu_1^{j,a',a''}, \ldots, \mu_K^{j,a',a''}) \leftarrow KSP(\mathbf{O}^{j,a'}, a', a'')$;
for $k = 1 : K$ **do**
  if $\psi_k^{j,a',a''}$ *is not a simple path* **then**
    break;
  Calculate the delay l_k of path $\psi_k^{j,a',a''}$ through $\mathbf{L}^{j,a'}$;
  if $l_k \leq t_j$ **then**
    return $\lambda_{ban}^{j,a',a''} \leftarrow \mu_k^{j,a',a''}$, $\psi^{j,a',a''} \leftarrow \psi_k^{j,a',a''}$;
$(\psi_L^{j,a',a''}, l_{\min}) \leftarrow DijkstraSP(\mathbf{L}^{j,a'}, a', a'')$;
if $l_{\min} \leq t_j$ **then**
  Calculate the bandwidth cost $\mu_L^{j,a',a''}$ of the path $\psi_L^{j,a',a''}$ through $\mathbf{O}^{j,a'}$;
  return $\lambda_{ban}^{j,a',a''} \leftarrow \mu_L^{j,a',a''}$, $\psi^{j,a',a''} \leftarrow \psi_L^{j,a',a''}$;
return $\lambda_{ban}^{j,a',a''} \leftarrow \infty$, $\psi^{j,a',a''} \leftarrow Null$;

η^{j,a_n} is the scheme that $f_1, f_2, \ldots, f_{j-1}$ are placed according to η^{j-1,a^*} and f_j is placed at node a_n (or node d). ψ^{j,a_n} is constructed by connecting ψ^{j-1,a^*} and ψ^{j,a^*,a_n} (or $\psi^{j,a^*,d}$). **Finally**, SDE constructs $\mathbf{c}^{j,a_n}$ and $\mathbf{B}^{j,a_n}$ by updating $\mathbf{c}^{j-1,a^*}$ and $\mathbf{B}^{j-1,a^*}$.

After establishing the *j*-th stage graph, if $\xi^{j,a_n} = \infty$ for $\forall a_n \in V_{ser}$, the establishment of the *j*-th stage graph is considered to be failed, and SDE rejects the source-destination pair (namely the user at the destination node) and terminates. If SDE successfully finishes establishing the last stage graph, the VNF placement and routing of this source-destination pair are respectively specified according to $\eta^{|F|+1,a^{**}}$ and $\psi^{|F|+1,a^{**}}$, and here $a^{**} = \arg\min_{a_n \in V_{ser}} \xi^{|F|+1,a_n}$. Then, SDE updates $\mathbf{P}$, $\mathbf{Q} = (\mathbf{Q}^{h_1}, \mathbf{Q}^{h_2}, \ldots, \mathbf{Q}^{h_{|F|+1}})$, $\mathbf{c}$, and $\mathbf{B}$ according to $\sigma^{|F|+1,a^{**}}$. The overall procedure of SDE is depicted in Algorithm 2.

Now we analyze the time complexity of SDE under the worst case. Constructing $\mathbf{O}^{j,a'}$ and $\mathbf{L}^{j,a'}$ needs $O(2|V|^2) = O(|V|^2)$ steps. In Algorithm 1, finding the K shortest paths in $\mathbf{O}^{j,a'}$ requires $O(|\hat{E}| + |V|\log|V| + K) = O(2|E| + |V|\log|V| + K) = O(|E| + |V|\log|V| + K)$ steps according to [44], while the shortest path in $\mathbf{L}^{j,a'}$ can be found by the Dijkstra algorithm [45] in $O(|V|^2)$ (note that $\mathbf{O}^{j,a'}$ and $\mathbf{L}^{j,a'}$ are both directed adjacency matrices). Since the number of links in the $\psi_k^{j,a',a''}$ and $\psi_L^{j,a',a''}$ is not larger than $|E|$, the time complexity of Algorithm 1 is $O(|E| + |V|\log|V| + K) + O(K|E|) + O(|V|^2) + O(|E|) = O(|V|^2 + K|E|)$. Constructing $\mathbf{c}^{j,a_n}$ and $\mathbf{B}^{j,a_n}$ needs $O(1 + |E|) = O(|E|)$ steps. Here note that when $|F| = 1$ (namely SFC length $|\hat{F}|$ is 3), lines 12-20 in Algorithm 2 will not be executed. Hence, the time complexity of SDE under the worst case is $O(|V_{ser}|(|V|^2 + K|E|))$ (when $|\hat{F}| = 3$) or $O(|F||V_{ser}|^2(|V|^2 + K|E|))$ (when $|\hat{F}| > 3$).

Algorithm 2: Source-Destination Embedding (SDE)

Input: $G = (V, E)$, $\mathbf{P}$, $\mathbf{Q} = (\mathbf{Q}^{h_1}, \mathbf{Q}^{h_2}, \ldots, \mathbf{Q}^{h_{|F|+1}})$, $\mathbf{c}$, $\mathbf{B}$, s, d, F, H, T, b, Γ.
Output: VNF placement strategy η and routing path ψ of the source-destination pair s and d, updated $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, $\mathbf{B}$.

1 Initialize $\mathbf{c}^{0,s} \leftarrow \mathbf{c}$, $\mathbf{B}^{0,s} \leftarrow \mathbf{B}$, $\eta^{0,s} \leftarrow Null$, $\psi^{0,s} \leftarrow Null$, $\xi^{0,s} \leftarrow 0$;
2 $\sigma^{0,s} \leftarrow (\mathbf{c}^{0,s}, \mathbf{B}^{0,s}, \eta^{0,s}, \psi^{0,s}, \xi^{0,s})$;
3 Construct $\mathbf{O}^{1,s}$ and $\mathbf{L}^{1,s}$ according to equations (25) and (26);
4 **for** $j = 1 : |F| + 1$ **do**
5   **for** $n = 1 : |V_{ser}|$ **do**
6   **if** $j = 1$ **then**
7   Calculate λ_{com}^{1,s,a_n} according to equation (24);
8   Calculate λ_{ban}^{1,s,a_n} and ψ^{1,s,a_n} through $\mathbf{O}^{1,s}$, $\mathbf{L}^{1,s}$, t_1 and Algorithm 1;
9   Calculate λ_{mig}^{1,a_n} according to equation (27);
10       $\lambda^{1,s,a_n} \leftarrow \lambda_{com}^{1,s,a_n} + \lambda_{ban}^{1,s,a_n} + \lambda_{mig}^{1,a_n}$, $\xi^{1,a_n} \leftarrow \lambda^{1,s,a_n}$;
11       $a^* \leftarrow s$, $\eta^{1,a_n} \leftarrow a_n$, $\psi^{1,a_n} \leftarrow \psi^{1,s,a_n}$;
12   **else if** $2 \leq j \leq |F|$ **then**
13   Calculate λ_{mig}^{j,a_n} according to equation (27);
14   **for** $m = 1 : |V_{ser}|$ **do**
15   Calculate $\lambda_{com}^{j,a_m,a_n}$ according to equation (24);
16   Calculate $\lambda_{ban}^{j,a_m,a_n}$ and ψ^{j,a_m,a_n} through $\mathbf{O}^{j,a_m}$, $\mathbf{L}^{j,a_m}$, t_j and Algorithm 1;
17         $\lambda^{j,a_m,a_n} \leftarrow \lambda_{com}^{j,a_m,a_n} + \lambda_{ban}^{j,a_m,a_n} + \lambda_{mig}^{j,a_n}$;
18       $\xi^{j,a_n} \leftarrow \min_{a_m \in V_{ser}} \{\xi^{j-1,a_m} + \lambda^{j,a_m,a_n}\}$;
19       $a^* \leftarrow \arg\min_{a_m \in V_{ser}} \{\xi^{j-1,a_m} + \lambda^{j,a_m,a_n}\}$;
20       $\eta^{j,a_n} \leftarrow (\eta^{j-1,a^*}, a_n)$, $\psi^{j,a_n} \leftarrow (\psi^{j-1,a^*}, \psi^{j,a^*,a_n})$;
21   **else if** $j = |F| + 1$ **then**
22   Calculate $\lambda_{ban}^{|F|+1,a_n,d}$ and $\psi^{|F|+1,a_n,d}$ through $\mathbf{O}^{|F|+1,a_n}$, $\mathbf{L}^{|F|+1,a_n}$, $t_{|F|+1}$ and Algorithm 1;
23       $\lambda^{|F|+1,a_n,d} \leftarrow \lambda_{ban}^{|F|+1,a_n,d}$;
24       $\xi^{|F|+1,a_n} \leftarrow \xi^{|F|,a_n} + \lambda^{|F|+1,a_n,d}$, $a^* \leftarrow a_n$;
25       $\eta^{|F|+1,a_n} \leftarrow (\eta^{|F|,a_n}, d)$;
26       $\psi^{|F|+1,a_n} \leftarrow (\psi^{|F|,a_n}, \psi^{|F|+1,a_n,d})$;
27   Construct $\mathbf{c}^{j,a_n}$ and $\mathbf{B}^{j,a_n}$ by updating $\mathbf{c}^{j-1,a^*}$ and $\mathbf{B}^{j-1,a^*}$;
28     $\sigma^{j,a_n} \leftarrow (\mathbf{c}^{j,a_n}, \mathbf{B}^{j,a_n}, \eta^{j,a_n}, \psi^{j,a_n}, \xi^{j,a_n})$;
29   **if** $1 \leq j \leq |F|$ **then**
30   Construct $\mathbf{O}^{j+1,a_n}$ and $\mathbf{L}^{j+1,a_n}$ according to equations (25) and (26);
31   **if** $\xi^{j,a_n} = \infty$ *for* $\forall a_n \in V_{ser}$ **then**
32   **return** $\eta \leftarrow Null$, $\psi \leftarrow Null$, $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, $\mathbf{B}$;
33 $a^{**} = \arg\min_{a_n \in V_{ser}} \xi^{|F|+1,a_n}$;
34 Update $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, $\mathbf{B}$ according to $\sigma^{|F|+1,a^{**}}$;
35 **return** $\eta \leftarrow \eta^{|F|+1,a^{**}}$, $\psi \leftarrow \psi^{|F|+1,a^{**}}$, $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, $\mathbf{B}$;

B. SDE-MSE

SDE-MSE specifies VNF placement and routing for each source-destination pair one by one with SDE. It is worth noting that, in SDE, the latter source-destination pair's VNF placement and routing scheme is affected by the previous pair's due to the record matrices. Hence, determining a proper order for all source-destination pairs is essential. SDE-MSE first finds central destination node d^* with the minimum sum of the distance to other destination nodes and then preferentially handles the destination node that is close to d^*. The distance can be

Algorithm 3: SDE-MSE

Input: $G = (V, E)$, $r = r_{MSE}$, $\mathbf{c}$, $\mathbf{B}$, $\mathbf{J}$.
Output: The VNF placement and routing schemes of all source-destination pairs.

Initialize $\mathbf{P}$, $\mathbf{Q} = (\mathbf{Q}^{h_1}, \mathbf{Q}^{h_2}, \ldots, \mathbf{Q}^{h_{|F|+1}})$;
$d^* \leftarrow \arg\min_{d \in D}\{\sum_{d' \in D} \pi_{d'd} | d \in D\}$;
$\tilde{D} \leftarrow$ Sort D according to the distance to d^* in ascending order;
for $d \in \tilde{D}$ **do**
    Use SDE (Algorithm 2) to decide VNF placement and routing for source-destination pair s and d;

either the minimum hop count or the minimum delay between two nodes (we can also consider both hop count and delay by weighted aggregation). The overall procedure of SDE-MSE is shown in Algorithm 3. In Algorithm 3, $\mathbf{J} = \{\pi_{uv}\}_{|V| \times |V|}$ is the pre-calculated distance matrix whose element π_{uv} is the minimum hop count or the delay from node u to node v.

Here we calculate the time complexity of SDE-MSE under the worst case. Determining the central destination node needs $O(|D|^2)$ steps and sorting D requires $O(|D| \log |D|)$ steps. SDE needs $O(|V_{ser}|(|V|^2 + K|E|))$ (when $|\hat{F}| = 3$) or $O(|F||V_{ser}|^2(|V|^2 + K|E|))$ (when $|\hat{F}| > 3$) steps. Since $|D| < |V|$, the time complexity of SDE-MSE under the worst case is $O(|D||V_{ser}|(|V|^2 + K|E|))$ (when $|\hat{F}| = 3$) or $O(|D||F||V_{ser}|^2(|V|^2 + K|E|))$ (when $|\hat{F}| > 3$).

C. SDE-MSR

Intuitively, the MSR problem can be solved by first releasing the SFC orchestration schemes of all existing users and then using SDE-MSE. Nevertheless, a tricky constraint in the MSR problem is to guarantee the preservation of all existing users, but SDE-MSE cannot ensure the admission of all users. Therefore, SDE-MSE can only solve the MSR problem when the residual network resources are fully sufficient and the delay constraint is fully loose. Here we propose SDE-MSR to tackle the MSR problem. SDE-MSR takes a conservative approach that releases one existing source-destination pair's scheme at a time and then applies SDE to deal with the source-destination pair. If the source-destination pair is admitted by SDE and overall resource consumption is reduced compared to the state before release, this source-destination pair is successfully adjusted, and SDE-MSR continues to handle the next source-destination pair. Otherwise, this source-destination pair's scheme reverts to the original one before release, and SDE-MSR continues to deal with the next source-destination pair. Like SDE-MSE, SDE-MSR makes the order of all source-destination pairs by first finding the central destination node d^* with the minimum sum of the distance to other destination nodes. However, on the contrary, SDE-MSR preferentially handles the destination node far from d^*. The overall procedure of SDE-MSR is summarized in Algorithm 4. Considering that SDE-MSR takes the conservative approach, we can add loops in the SDE-MSR (line 4) to further improve performance.

Now we analyze the time complexity of SDE-MSR under the worst case. Releasing the VNF placement and routing scheme of one source-destination pair and updating $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, and $\mathbf{B}$ requires $O((|F| + 1)|E|) = O(|F||E|)$ steps.

Algorithm 4: SDE-MSR

Input: $G = (V, E)$, $r = r_{MSR}$, $\mathbf{c}$, $\mathbf{B}$, $\mathbf{J}$.
Output: The VNF placement and routing schemes of all source-destination pairs.

1 Initialize $\mathbf{P}$, $\mathbf{Q} = (\mathbf{Q}^{h_1}, \mathbf{Q}^{h_2}, \ldots, \mathbf{Q}^{h_{|F|+1}})$;
2 $d^* \leftarrow \arg\min_{d \in D}\{\sum_{d' \in D} \pi_{d'd} | d \in D\}$;
3 $\tilde{D} \leftarrow$ Sort D according to the distance to d^* in descending order;
4 **for** $loop = 1 : \varpi$ **do**
5   **for** $d \in \tilde{D}$ **do**
6         $\mathbf{P}^* \leftarrow \mathbf{P}$, $\mathbf{Q}^* \leftarrow \mathbf{Q}$, $\mathbf{c}^* \leftarrow \mathbf{c}$, $\mathbf{B}^* \leftarrow \mathbf{B}$;
7   Release the VNF placement and routing scheme of the source-destination pair s and d and update $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, $\mathbf{B}$;
8   Use SDE (Algorithm 2) to decide VNF placement and routing for source-destination pair s and d;
9   **if** *d is rejected or overall resource consumption is not reduced compared with the state before release* **then**
10   The scheme of source-destination pair s and d reverts to the original one before release;
11             $\mathbf{P} \leftarrow \mathbf{P}^*$, $\mathbf{Q} \leftarrow \mathbf{Q}^*$, $\mathbf{c} \leftarrow \mathbf{c}^*$, $\mathbf{B} \leftarrow \mathbf{B}^*$;

Algorithm 5: SDE-MSEP

Input: $G = (V, E)$, $r = r_{MSEP}$, $\mathbf{c}$, $\mathbf{B}$, $\mathbf{J}$.
Output: The VNF placement and routing schemes of all source-destination pairs.

Use SDE-MSR (Algorithm 4) to deal with D_{eu};
Use existing $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, $\mathbf{B}$ and SDE-MSE (Algorithm 3) to deal with D_{nu};

SDE requires $O(|V_{ser}|(|V|^2 + K|E|))$ (when $|\hat{F}| = 3$) or $O(|F||V_{ser}|^2(|V|^2 + K|E|))$ (when $|\hat{F}| > 3$) steps. Hence, the time complexity of SDE-MSR under the worst case is $O(\varpi|D||V_{ser}|(|V|^2 + K|E|))$ (when $|\hat{F}| = 3$) or $O(\varpi|D||F||V_{ser}|^2(|V|^2 + K|E|))$ (when $|\hat{F}| > 3$). Here ϖ is the loop number.

D. SDE-MSEP

SDE-MSEP can be considered as the combination of SDE-MSR and SDE-MSE because it first applies SDE-MSR to deal with existing users and then uses SDE-MSE to handle new users. Here note that, in SDE-MSEP, SDE-MSE continues to use the $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{c}$, and $\mathbf{B}$ output by SDE-MSR rather than initializing new ones. The overall procedure of SDE-MSEP is depicted in Algorithm 5.

Here we calculate the time complexity of SDE-MSEP under the worst case. Since SDE-MSEP combines SDE-MSE and SDE-MSR, the time complexity of SDE-MSEP under the worst case is $O((\varpi|D_{eu}| + |D_{nu}|)|V_{ser}|(|V|^2 + K|E|))$ (when $|\hat{F}| = 3$) or $O((\varpi|D_{eu}| + |D_{nu}|)|F||V_{ser}|^2(|V|^2 + K|E|))$ (when $|\hat{F}| > 3$).

VI. Simulation Results

In this section, we present simulation results to evaluate the performance of the proposed algorithms SDE-MSE, SDE-MSR, and SDE-MSEP.

A. Simulation Setup

1) Simulated Network: As in [14], we apply the real-world network of PalmettoNet [46] as the physical network topology. PalmettoNet is a backbone network with 45 nodes. Still following [14], we also use the ER random graph [47] to synthetically generate network topologies to evaluate the performance of our algorithms under different network sizes. The size of the ER random network topology is set in the range of [50, 125]. We randomly select 30% of nodes as forwarding nodes while the others are service nodes. The residual computing capacity of each service node is generated following the uniform distribution $U(10, 50)$. The link delay is set to be proportional to the Euclidean distance between the pair of nodes [18], and the residual bandwidth capacity of each link is generated following the uniform distribution $U(100, 500)$. We set that the network supports 5 kinds of VNFs [30], and the required computing resource of each VNF follows the uniform distribution $U(10, 12)$.

2) MSO Request: The source and destination nodes are randomly selected from the network ($s \notin D$). We set SFC length $|\hat{F}|$ ($|\hat{F}| = |F| + 2$) from 3 to 7 [30], and each VNF in the SFC is randomly selected from the VNF pool. Each virtual link delay constraint is proportional to the average minimum delay as in equation (29), where π_{uv}^{delay} is the minimum delay between node u and node v in the network and ϑ_t is the tuning parameter. We set $\vartheta_t = 1$ by default. Bandwidth requirement b is set to follow the uniform distribution $U(100, 120)$. For MSR and MSEP requests, we randomly generate a feasible scheme as Γ. For MSE and MSEP requests, we set per new user admission revenue proportional to SFC length and the average minimum hop count as equation (30) to balance the revenues and the costs. In equation (30), ϑ_κ, κ_{bas}, and π_{uv}^{hop} are respectively the tuning parameter, the basic admission revenue of each new user, and the minimum hop count between node u and node v in the network. We set $\vartheta_\kappa = 1$ and $\kappa_{bas} = 120$.

$$t_j = \vartheta_t \cdot \frac{1}{|V|^2 - |V|} \cdot \sum_{u \in V} \sum_{v \in V, v \neq u} \pi_{uv}^{delay}, \ \forall t_j \in T \quad (29)$$

$$\kappa = \vartheta_\kappa \cdot \kappa_{bas} \cdot |\hat{F}| \cdot \frac{1}{|V|^2 - |V|} \cdot \sum_{u \in V} \sum_{v \in V, v \neq u} \pi_{uv}^{hop}. \quad (30)$$

3) Cost Factors: The cost of unit link bandwidth resource is set as $\beta = 1$. Unit node computing resource cost α and migration cost coefficient $\gamma_{u,v}^f$ are set to be respectively proportional to the average minimum hop count [14] and the minimum hop count between two nodes to keep the computing cost, bandwidth cost, and migration cost at the same quantitative level. In detail, α and $\gamma_{u,v}^f$ are respectively set according to equations (31) and (32), where ϑ_α and ϑ_γ are corresponding tuning parameters set as $\vartheta_\alpha = 12$ and $\vartheta_\gamma = 10$. We assume that a VNF with a higher computing resource requirement has a larger software size, thereby incurring a higher migration cost. Therefore, we make $\gamma_{u,v}^f$ also proportional to ω_f as in equation (32).

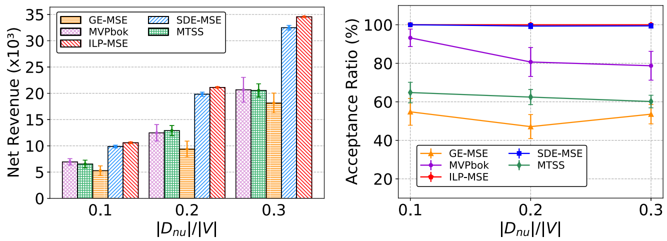
$$\alpha = \vartheta_\alpha \cdot \frac{1}{|V|^2 - |V|} \cdot \sum_{u \in V} \sum_{v \in V, v \neq u} \pi_{uv}^{hop} \quad (31)$$

$$\gamma_{u,v}^f = \vartheta_\gamma \cdot \omega_f \cdot \pi_{uv}^{hop}. \quad (32)$$

4) Benchmark Algorithms: To our knowledge, this is the first work to study MSR and MSEP problems, and there are no existing algorithms that deal with the two problems. Therefore, we first devise the greedy embedding (GE) algorithm, a variant of the SDE algorithm, and obtain GE-MSE, GE-MSR, and GE-MSEP by replacing the SDE part in SDE-MSE, SDE-MSR, and SDE-MSEP with GE. We then use GE-MSE, GE-MSR, and GE-MSEP as the benchmarks of SDE-MSE, SDE-MSR, and SDE-MSEP, respectively. GE is the same as SDE except that GE greedily determines the service node placing f_j after establishing the j-th stage graph according to the accumulative cost rather than deciding after constructing all stage graphs through a series of assumptions like SDE.

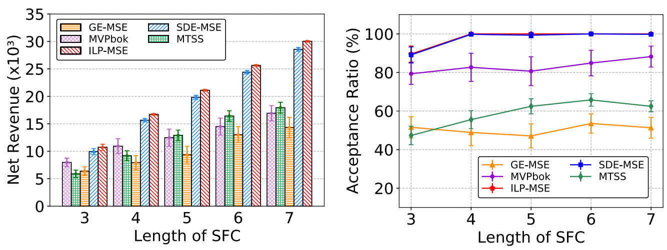
In addition, for the MSE problem, we also compare our algorithm SDE-MSE with the following two existing heuristic algorithms proposed in [16]: Multicast VNF Placement based on K-Shortest Path (MVPboK) and Multicast Tree with Source Selection (MTSS). Specifically, MVPboK first determines the central destination node based on the hop count. Then it finds the KSP from the source node to the central destination node and selects an appropriate path in which all VNFs can be successfully placed. After that, for each destination node, MVPboK finds KSP from the source node to it and selects the path in which all VNFs can be successfully placed while VNF reuse can be maximized. MTSS first determines the central destination node in the same way and then places all the VNFs of the SFC along the path from the source node to the central node. Afterward, MTSS connects the remaining destination nodes by finding the shortest paths from the node placing the last VNF to them. A common feature of MVPboK and MTSS is that both of them place VNFs after obtaining a particular simple path. Since the authors in [16] do not demonstrate how to place VNFs after determining a path, in our simulations, we apply the dynamic programming to place VNFs [48] on a determined path in MVPboK and MTSS.

5) Evaluation Method: Corresponding to the three different kinds of MSO problems, namely MSE, MSR, and MSEP problems, the whole simulation is composed of three parts: the evaluations of SDE-MSE, SDE-MSR, and SDE-MSEP. In each part, we first evaluate the impacts of destination ratio/constitution, SFC length, and delay constraint degree on performance. Here we compare our algorithms against the benchmark algorithms and the optimal solution in the real-world network topology of PalmettoNet. The optimal solution is obtained by solving the corresponding ILP with the CPLEX solver [49]. Specifically, for the MSE problem, we compare SDE-MSE against GE-MSE, MVPboK, MTSS, and ILP-MSE with respect to total net revenue and acceptance ratio under different new destination ratios, SFC lengths, and delay constraint degrees. For the MSR problem, we compare SDE-MSR with GE-MSR and ILP-MSR in terms of total net revenue under different existing destination ratios, SFC lengths, and delay constraint degrees. The destination ratio in MSE and MSR requests is set as $|D|/|V| = 0.1$ or $|D|/|V| = 0.2$ or $|D|/|V| = 0.3$ [14]. For the MSEP problem, we compare SDE-MSEP against GE-MSEP and ILP-MSEP with respect to



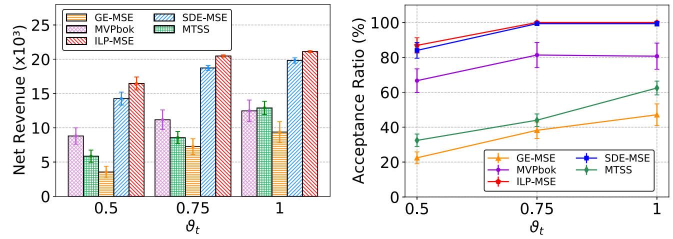
(a) Net revenue (b) Acceptance ratio

Fig. 4. Performance with the increase of the new destination ratio.



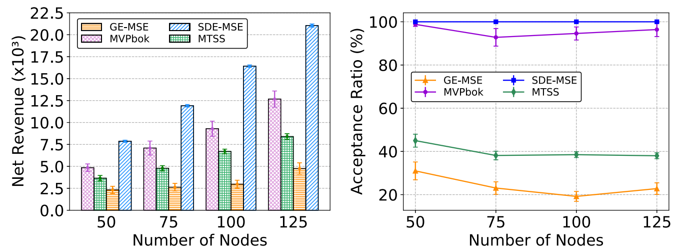
(a) Net revenue (b) Acceptance ratio

Fig. 5. Performance with the increase of SFC length.



(a) Net revenue (b) Acceptance ratio

Fig. 6. Performance under different delay constraint degrees.



(a) Net revenue (b) Acceptance ratio

Fig. 7. Performance under different network sizes.

total net revenue and acceptance ratio under different destination constitutions, SFC lengths, and delay constraint degrees. The constitution of destination in MSEP request is set as $|D_{nu}|/|D_{eu}| = 0.5$ or $|D_{nu}|/|D_{eu}| = 1$ or $|D_{nu}|/|D_{eu}| = 2$ (we keep $(|D_{nu}| + |D_{eu}|)/|V| = 0.3$). Then in each part, to evaluate the impact of network size on performance, we compare our algorithm against the benchmark algorithms under the different sizes of ER random network topology. Here we do not compare with the optimal solution as the CPLEX solver takes too much time to obtain the optimal solution. We repeat each kind of simulation 50 rounds under the same parameter setting and report the results with 95% confidence interval. All simulations are conducted in a laptop with Intel Core i5-7300HQ (2.50GHz) processors and 8GB of RAM. Moreover, we set $K = 3$ in KSP and set $\mathbf{J} = \{\pi_{uv} = \pi_{uv}^{delay}\}_{|V|\times|V|}$ and $\varpi = 1$ in SDE-MSR.

B. Evaluation of SDE-MSE

1) The Impact of New Destination Ratio: The new destination ratio is defined as the number of new user destination nodes in the MSE request to the total number of network nodes, namely $|D_{nu}|/|V|$. To evaluate the impact of the new destination ratio on performance, we fix the length of SFC with 5. Fig. 4 shows the net revenue and acceptance ratio results with the increasing new destination ratio. Here the acceptance ratio is the number of admitted new user destination nodes to the number of all new user destination nodes in the MSE request. As shown in the figure, the achieved net revenues by all methods grow with the increase of the new destination ratio because a higher new destination ratio means more chances to admit new users. We can also observe in the figure that SDE-MSE outperforms the benchmark algorithms in both net revenue and acceptance ratio and keeps stable and near-optimal performance under different new destination ratios. The net revenue and acceptance ratio obtained by ILP-MSE are just 1.07 and 1.004 times those by SDE-MSE on average, as illustrated in Fig. 4(a) and Fig. 4(b), respectively.

For the benchmark algorithms, GE-MSE achieves the worst performance in both net revenue and acceptance ratio, as GE in GE-MSE greedily establishes each stage and reduces the search space. Such a feature makes GE-MSE easily trapped in local optimum and harms delay optimization. MVPboK has a relatively high acceptance ratio but low net revenue, which means it results in high resource costs. Like SDE-MSE, MVPboK deals with each source-destination pair one by one, which is beneficial to handle delay constraint and thus increases the acceptance ratio. However, MVPboK does not fully reuse the links in multicast routing, and this incurs a high bandwidth cost. In contrast, MTSS achieves a relatively low acceptance ratio but high net revenue. The reason is that MTSS restricts the replication point in multicast routing to occur after the flow processing of all VNFs. Such restriction reduces feasible solution space and is detrimental to satisfying virtual link delay constraint, leading to a low acceptance ratio. However, it also decreases routing branches and thereby saves resource costs. Unlike these algorithms, SDE-MSE applies the record matrices to maximize the reuse of VNFs and links as possible. At the same time, the SDE in SDE-MSE enlarges the search space through a series of assumptions to prevent SDE-MSE from being easily trapped in the local optimum. On average, the net revenue and acceptance ratio achieved by SDE-MSE are 92.57% and 92.98% higher than GE-MSE's, 52.66% and 18.92% higher than MVPboK's, and 54.21% and 59.57% higher than MTSS's respectively, as shown in Fig. 4(a) and Fig. 4(b).

2) The Impact of SFC Length: We restrict the new destination ratio to 0.2 and evaluate the impact of SFC length on performance. The net revenue and acceptance ratio variations as the increase of SFC length are illustrated in Fig. 5. We can observe in the figure that the net revenues of all methods also grow with the increase of SFC length because a longer SFC can bring more revenue according to equation (30). Still, SDE-MSE performs much better than the benchmark algorithms and achieves near-optimal performance under different SFC lengths. The net revenue and acceptance ratio derived by ILP-MSE are 1.06 and 1.003 times SDE-MSE's on average, as shown in Fig. 5(a) and Fig. 5(b), respectively.

By comparison, GE-MSE performs unstably in both the net revenue and acceptance ratio due to its greedy feature. Similar to the previous evaluation, MVPboK achieves a relatively high acceptance ratio but low net revenue, while MTSS results in a relatively low acceptance ratio but high net revenue. On average, our SDE-MSE can increase the net revenue and acceptance ratio by 90.36% and 93.78% against GE-MSE, 52.79% and 17.41% against MVPboK, and 60.2% and 67.75% against MTSS respectively, as illustrated in Fig. 5(a) and Fig. 5(b).

An interesting observation is made in Fig. 5(b) the acceptance ratios achieved by SDE-MSE, MVPboK, MTSS, and ILP-MSE are relatively low when the SFC length is 3. This is because, in a fixed scale network, a virtual link of a shorter SFC usually requires more links to construct. Obviously, this is not conducive to satisfying virtual link delay constraint and hence reduces the acceptance ratio.

3) The Impact of Delay Constraint Degree: To evaluate the impact of the delay constraint degree on performance, we carry simulations under different degrees of delay constraint by adjusting the tuning parameter ϑ_t in equation (29). The new destination ratio and SFC length are respectively set as 0.2 and 5, and the result is shown in Fig. 6. As expected, the net revenues achieved by all methods grow with the increase of ϑ_t because larger ϑ_t indicates the looser delay constraint. Even under the strict delay constraint ($\vartheta_t = 0.5$), SDE-MSE still achieves good and near-optimal performance in both net revenue and acceptance ratio. When $\vartheta_t = 0.5$, the net revenue and acceptance ratio obtained by ILP-MSE are just 1.16 and 1.03 times SDE-MSE's in Fig. 6(a) and Fig. 6(b), respectively.

As mentioned before, like SDE-MSE, MVPboK handles each source-destination pair one by one and this is conducive to satisfying the delay constraint. Therefore, MVPboK still achieves a relatively high acceptance ratio when the delay constraint is strict ($\vartheta_t = 0.5$). By comparison, due to the replication point restriction, the acceptance ratio resulting from MTSS is relatively low when ϑ_t is small.

4) The Impact of Network Size: We compare SDE-MSE with the benchmark algorithms under the different sizes of ER random network topology to evaluate the impact of network size. We fix the new destination ratio with 0.2 and the SFC length with 5. Fig. 7 shows the net revenue and acceptance ratio results. As we observe in the figure, the net revenues resulting from all the algorithms increase with the growth of network size because admitting a new user in a larger network brings more revenue according to equation (30). Among the algorithms, SDE-MSE goes beyond the benchmark algorithms and keeps stable performance regardless of the network size. By comparison, the benchmark algorithms achieve much lower net revenues than SDE-MSE, and the gaps become larger with the growth of network size. Besides, the acceptance ratios from the benchmark algorithms slightly decrease when the network size grows.

TABLE III
RUNTIME COMPARED WITH ILP-MSE

Network Topology	**PalmettoNet (45Nodes)**		
New Destination Ratio ($\lvert\hat{F}\rvert=5$)	0.1	0.2	0.3
SDE-MSE	6.5s	11.9s	16.5s
ILP-MSE	233.0s	617.7s	1304.9s
SFC Length ($\lvert D_{nu}\rvert/\lvert V\rvert=0.2$)	3	5	7
SDE-MSE	0.41s	11.9s	26s
ILP-MSE	202.2s	617.7s	1205.6s

5) Runtime Compared to ILP-MSE: In Table III, we attach the average runtime of SDE-MSE compared with ILP-MSE's. We can see in the table that the runtime of SDE-MSE is much less than ILP-MSE's, and their gap increases rapidly when the problem size grows. Besides, as analyzed before, the time complexity of SDE is low for the 3-length SFC, and hence SDE-MSE has short runtime when the SFC length is 3 as shown in the table.

C. Evaluation of SDE-MSR

1) The Impact of Existing Destination Ratio: We define the existing destination ratio as the number of existing user destination nodes in the MSR request to the total number of network nodes, namely $|D_{eu}|/|V|$. We restrict the SFC length to 5 and evaluate the impact of the existing destination ratio on performance. Fig. 8 presents the net revenue result as the increase of the existing destination ratio. We can observe in the figure that the net revenues of all methods grow when the existing destination ratio increases, as a larger existing destination ratio indicates more opportunities to save the network resources through adjustment. Besides, as shown in the figure, the net revenue derived by ILP-MSR is 1.34 times SDE-MSR's on average. SDE-MSR outperforms GE-MSR, and the net revenue achieved by SDE-MSR is 16.58% higher than that by GE-MSR on average, as shown in Fig. 8.

Compared with the previous simulation part, we can find that the performance gap between SDE-MSR and ILP-MSR is larger than that between SDE-MSE and ILP-MSE. The reason is that although the conservative approach taken by SDE-MSR can ensure the preservation of all existing users, it also increases the possibility of being trapped in the local optimum. In this regard, as already mentioned, we can add loops for SDE-MSR to further improve performance though this will increase the runtime.

2) The Impact of SFC Length: To evaluate the impact of SFC length, we set the existing destination ratio as 0.2. The net revenue variations of the three methods with the increasing SFC length are illustrated in Fig. 9. As shown in the figure, the net revenues obtained by all methods grow when the SFC length increases. The reason for this phenomenon is that a longer SFC has more VNFs and hence brings more chances to save the node computing resource through VNF adjustment.

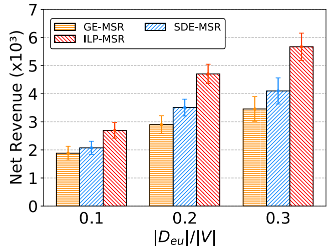


Fig. 8. Net revenue with the increase of the existing destination ratio.

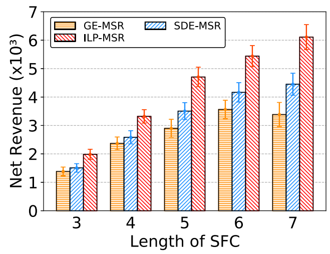


Fig. 9. Net revenue with the increase of SFC length.

The net revenue resulting from ILP-MSR is 1.31 times that from SDE-MSR on average in Fig. 9. Still, SDE-MSR goes beyond GE-MSR in net revenue, and the gap between them becomes larger with the increase of SFC length. On average, SDE-MSR can increase net revenue by 13.02% against GE-MSR in Fig. 9.

3) The Impact of Delay Constraint Degree: We compare SDE-MSR with ILP-MSR and GE-MSR under different degrees of delay constraint to evaluate the impact of delay constraint degree on performance. The existing destination ratio and SFC length are respectively fixed at 0.2 and 5. Fig. 10 presents the changing trend of net revenue as the increase of ϑ_t. We can see that when the delay constraint is strict, namely ϑ_t is small, all three methods achieve relatively small net revenues. The reason behind this is that the strict delay constraint reduces the adjustable space of the existing scheme, leading to fewer resource savings. When $\vartheta_t = 0.5$, the net revenue obtained by ILP-MSR is 1.31 times that by SDE-MSR, as shown in Fig. 10.

4) The Impact of Network Size: To evaluate the impact of network size on performance, we compare SDE-MSR against GE-MSR under the different sizes of ER random network topology. We restrict the existing destination ratio to 0.2 and the SFC length to 5, and the result is illustrated in Fig. 11. We can observe that the net revenues achieved by SDE-MSR and GE-MSR grow with the increase of network size. The reason for this phenomenon is that a larger network usually leads to a longer flow path and thus provides more opportunities to save the link bandwidth resource through routing adjustment. SDE-MSR outperforms GE-MSR in net revenue, and the gap between them becomes larger when the network size grows, as shown in Fig. 11.

5) Runtime Compared to ILP-MSR: Table IV presents the comparison of the average runtime between SDE-MSR and ILP-MSR. As shown in the table, the runtime of SDE-MSR is several orders of magnitude less than ILP-MSR's. Similar to

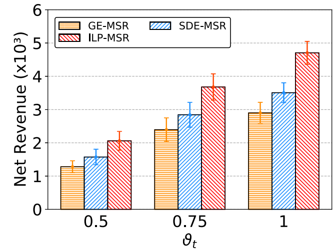


Fig. 10. Net revenue under different delay constraint degrees.

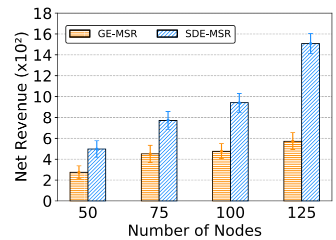


Fig. 11. Net revenue under different network sizes.

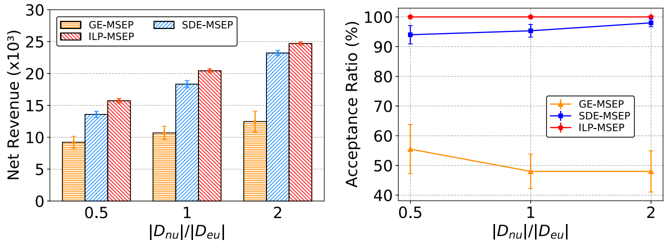
TABLE IV
RUNTIME COMPARED WITH ILP-MSR

Network Topology	**PalmettoNet (45Nodes)**		
Existing Destination Ratio ($|\hat{F}|=5$)	0.1	0.2	0.3
SDE-MSR	5.2s	6.7s	7s
ILP-MSR	251.4s	589.5s	1480.6s
SFC Length ($|D_{eu}|/|V|=0.2$)	3	5	7
SDE-MSR	0.37s	6.7s	7.1s
ILP-MSR	192.9s	589.5s	1212.7s

SDE-MSE, SDE-MSR results in a short runtime when the SFC length is 3 as SDE in SDE-MSR has a low time complexity for the 3-length SFC. By comparing Table III and Table IV, an interesting observation is made that SDE-MSR's runtime is less than SDE-MSE's. In SDE-MSR, we take the conservative approach that releases one source-destination pair's scheme of VNF placement and routing at a time and then applies SDE for adjustment. Since SDE tends to look for the VNF placement and routing scheme for one source-destination pair that can maximize the reuse of VNFs and links as possible, SDE is easily affected by the existing embedding paths of other source-destination pairs. When the existing embedding paths are highly distributed, the routing path found by SDE may be difficult to satisfy virtual delay constraint. Therefore, the probability of early termination of SDE (see lines 31-32 in Algorithm 2) is higher in SDE-MSR than in SDE-MSE, leading to less runtime.

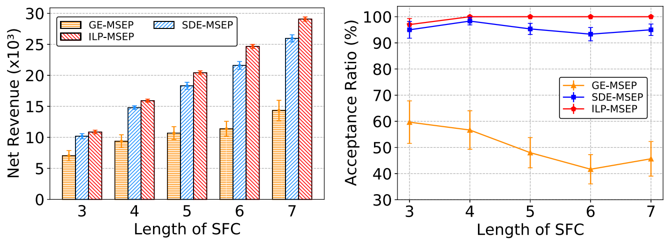
D. Evaluation of SDE-MSEP

1) The Impact of Destination Constitution: The destination constitution is defined as the number of new user destination nodes to the number of existing user destination nodes in the MSEP request, namely $|D_{nu}|/|D_{eu}|$. In the whole evaluation of SDE-MSEP, we keep $(|D_{nu}| + |D_{eu}|)/|V| = 0.3$. To evaluate the impact of the destination constitution on



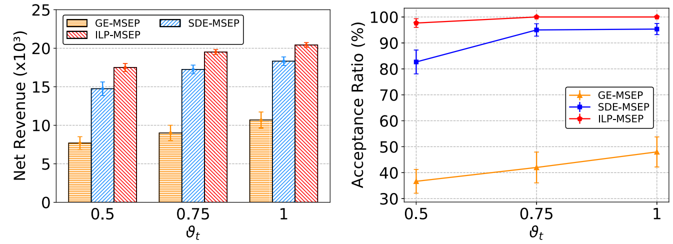
(a) Net revenue
(b) Acceptance ratio

Fig. 12. Performance with the increase of the destination constitution.



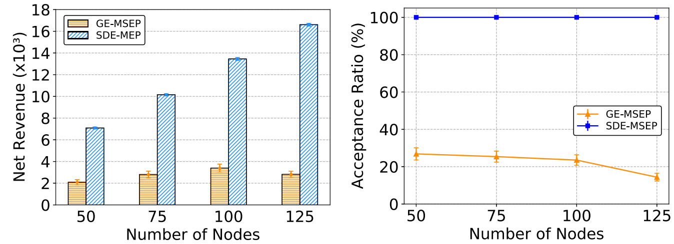
(a) Net revenue
(b) Acceptance ratio

Fig. 13. Performance with the increase of SFC length.



(a) Net revenue
(b) Acceptance ratio

Fig. 14. Performance under different delay constraint degrees.



(a) Net revenue
(b) Acceptance ratio

Fig. 15. Performance under different network sizes.

performance, we fix the length of SFC with 5. Fig. 12 presents the net revenue and acceptance ratio results as the increase of the destination constitution. Note that the acceptance ratio here is the number of admitted new user destination nodes to the number of all new user destination nodes in the MSEP request. As depicted in the figure, the net revenues of the three methods grow when the destination constitution increases because admitting new users is more conducive to revenue compared with adjustment. SDE-MSEP outperforms the benchmark algorithm and achieves near-optimal performance in both net revenue and acceptance ratio. On average, the net revenue and acceptance ratio obtained by ILP-MSEP are just 1.11 and 1.04 times those by SDE-MSEP in Fig. 12(a) and Fig. 12(b), respectively. Compared with GE-MSEP, SDE-MSEP can respectively increase the net revenue and acceptance ratio by 68.38% and 90.72% against GE-MSEP on average, as shown in Fig. 12(a) and Fig. 12(b).

An observation is made in Fig. 12(a) that the net revenue gap between SDE-MSEP and ILP-MSEP narrows when the destination constitution increases. SDE-MSEP is the combination of SDE-MSR and SDE-MSE, and through the previous evaluations, we can find that SDE-MSE can achieve more near-optimal performance than SDE-MSR. Since a larger destination constitution means more new user destination nodes and fewer existing user destination nodes, the performance of SDE-MSEP gets closer to the optimum with the increase of destination constitution as SDE-MSE plays more roles.

2) The Impact of SFC Length: We restrict the destination constitution to 1 and evaluate the impact of SFC length on performance. The changing trends of net revenue and acceptance ratio as the increase of SFC length are illustrated in Fig. 13. As shown in the figure, the net revenues achieved by all methods grow with the increase of SFC length because a longer SFC brings more admitting revenue and opportunities of VNF adjustment. SDE-MSEP still goes beyond the benchmark algorithm and performs near-optimal performance. The net revenue and acceptance ratio derived by ILP-MSEP are 1.1 and 1.04 times those by SDE-MSEP in Fig. 13(a) and Fig. 13(b), respectively. When compared with GE-MSEP, SDE-MSEP's net revenue and acceptance ratio are respectively 68.94% and 92.68% higher than GE-MSEP's, as illustrated in Fig. 13(a) and Fig. 13(b).

We can observe in Fig. 13(b) that the average acceptance ratio achieved by SDE-MSEP fluctuates when the SFC length varies. The mechanism of SDE-MSEP is to first use SDE-MSR and then use SDE-MSE. Considering that the SDE in SDE-MSE is affected by the embedding paths of existing source-destination pairs, the readjustment result of SDE-MSR will have a certain degree of influence on SDE-MSE. Under the fixed destination constitution and delay constraint degree, such influence is obvious, leading to the fluctuation in the acceptance ratio.

3) The Impact of Delay Constraint Degree: To evaluate the impact of delay constraint degree, we fixed the destination constitution with 1 and the SFC length with 5. Fig. 14 presents the net revenue and acceptance ratio results with the increase of ϑ_t. As expected, the net revenues and acceptance ratios resulting from the three methods increase when ϑ_t becomes larger. SDE-MSEP still keeps near-optimal performance even though the delay constraint is strict. When $\vartheta_t = 0.5$, the net revenue and acceptance ratio achieved by ILP-MSEP are just 1.19 and 1.18 times those by SDE-MSEP, as shown in Fig. 14(a) and Fig. 14(b), respectively.

4) The Impact of Network Size: We compare SDE-MSEP against GE-MSEP under the different sizes of ER random network topology to evaluate the impact of network size on performance. We set the destination constitution to 1 and the

TABLE V
RUNTIME COMPARED WITH ILP-MSEP

Network Topology	**PalmettoNet (45Nodes)**		
Destination Constitution ($\lvert\hat{F}\rvert=5$)	0.5	1	2
SDE-MSEP	11.9s	14.5s	16s
ILP-MSEP	1407.4s	1290.1s	1241s
SFC Length ($\lvert D_{nu}\rvert/\lvert D_{eu}\rvert=1$)	3	5	7
SDE-MSEP	0.68s	14.5s	20.6s
ILP-MSEP	419.3s	1290.1s	3819.7s

SFC length to 5. Fig. 15 shows the net revenue and acceptance ratio results. From the figure, we can see that the net revenues obtained by SDE-MSEP and GE-MSEP grow with the increase of network size, as admitting new users and enforcing adjustments in a larger network bring more revenue. SDE-MSEP performs much better than GE-MSEP in both the net revenue and acceptance ratio, and the gaps become larger when the network size grows.

5) Runtime Compared to ILP-MSEP: In Table V, we present the average runtime of SDE-MSEP compared with ILP-MSEP's. We can see in the table that SDE-MSEP results in much less runtime than ILP-MSEP. Likewise, the runtime of SDE-MSEP is short when the SFC length is 3 due to the low complexity of SDE for the 3-length SFC. Besides, we can find that SDE-MSEP's runtime increases when the destination constitution becomes larger. SDE-MSEP consists of SDE-MSE and SDE-MSR, and through the previous evaluations, we find that SDE-MSE has longer runtime than SDE-MSR under a similar setting. Therefore, SDE-MSE plays more roles when the destination constitution is larger, leading to more runtime of SDE-MSEP.

VII. CONCLUSION AND FUTURE WORK

In this paper, we study the MSO in SDN/NFV-enabled networks, including MSE, MSR, and MSEP, with joint consideration of the flexible replication, resource limitation, and source-destination delay constraint to better adapt to the features of multicast service like multicast routing and user fluidity. We first define and formulate MSE, MSR, and MSEP problems and then propose three heuristic algorithms, SDE-MSE, SDE-MSR, and SDE-MSEP, to solve the above three kinds of MSO problems respectively. Extensive simulations are conducted to evaluate the proposed algorithms, and simulation results indicate that our algorithms go beyond the benchmark algorithms and achieve near-optimal performance.

We believe that MSO (especially MSR and MSEP) is worth further research. We intend to investigate MSO considering additional optimization factors or under different network scenarios. Besides, as mentioned before, a tricky constraint for utilizing heuristic methods to solve MSR and MSEP problems is ensuring the preservation of existing users. In SDE-MSR and SDE-MSEP, we take a conservative approach to guarantee the above constraint. Nevertheless, this comes at the expense of a certain performance. Therefore, we are also interested in investigating better solutions for MSR and MSEP problems.

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